Transcript: The Thermal Pure Higgsino at 1.0–1.2 TeV: Relic Mass, Direct-Detection Limits and the LZ 248 keV Event

HEP-Genesis-Agent session record
Project: ramp-research-higgsino-dd

2026-10-22

The conversation verbatim, generated mechanically from the session record.

A Conversation transcript

The exchange verbatim (assistant text as shown to the user; tool calls summarized by name — see the tool-call log for arguments and results).

User (2026-10-16 11:03:59)

Survey recent arXiv work on the higgsino thermal relic target (1.0-1.2 TeV). Use the inelastic-DM tools to compute the relic density at several masses; compare with LZ/XENONnT sensitivities and the Planck abundance. Where does the higgsino sit relative to current limits?

Assistant (research mode, 2026-10-16 11:03:59)

User (2026-10-16 11:04:37)

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Assistant (research mode, 2026-10-16 11:04:37)

Plan:

1. (closed) Establish the pure-higgsino thermal relic density as a function of mass (about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV), with and without Sommerfeld enhancement, and find the mass at which Omega h^2 equals the Planck 2018 value of 0.120 +/- 0.001.
2. (closed) Place the pure-higgsino elastic spin-independent cross section at about 1.0-1.2 TeV against the LZ 2024 (2410.17036) and XENONnT (2303.14729) limits at the same mass, and against the xenon neutrino fog.
3. (closed) Assess the routes by which a higgsino becomes visible to direct detection: the mass splittings and the inelastic Z-exchange threshold, LZ inelastic constraints (2609.09830), mixing-induced Higgs couplings in the (mu, M1/M2, tan beta) plane (2606.31013), and the (Omega h^2/0.12) rescaling for sub-thermal higgsinos.

4. (blocked) Pin down the pure-higgsino thermal mass from at least two independent published calculations, including recent work with bound-state formation, and quote the Planck 2018 Omega_c h^2 verbatim. Confirm which mass the 1411.6930 pure-higgsino values (0.126/0.127) refer to.

5. (blocked) Replace the linear 40 GeV extrapolation with direct reads of the LZ 2024 and XENONnT SI limits at about 1 TeV, a numeric xenon neutrino-fog value at about 1 TeV, and one next-generation projection (XLZD/DARWIN). Correct the f-008 arithmetic and quote Chen and Hill verbatim.

6. (closed) Verify 2606.31013 and 2609.09830 with metadata, and trace the contested f-002 and f-003 numbers to quoted excerpts: Eq. 13 (356 MeV), the f_q formula, Eq. 38, the Table VII NLO rows, and the tree-level delta_m0 formula (Eq. 9 / Ref. [28]).

7. (open) Identify Ref. [28] of arXiv:2609.09830 and quote its Eq. (9) for the neutral splitting

8. (open) Add a sourced statement on indirect-detection limits for TeV higgsinos

Tool calls: `list_agent_files`, `list_input_files`, `list_datasets`, `list_files`, `get_dispatch`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `get_arxiv_metadata`, `search_arxiv_abstract` `search_arxiv_abstracts`, `search_arxiv_abstracts`, `list_input_files`, `download_arxiv_full_text`, `download_arxiv_full_text`, `search_arxiv_abstracts`, `download_arxiv_full_text`, `search_arxiv_abstra` `download_arxiv_full_text`, `download_arxiv_full_text`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `search_arxiv_abstracts`, `read_arxiv_chunk`, `get_paper_figures`, `search_arxiv_abstracts`, `read_arxiv_chunk`, `search_arxiv_abstracts`, `get_arxiv_metadata`, `read_arxiv_chunk`, `download_arxiv_f` `download_arxiv_full_text`, `post_finding`, `post_finding`, `post_finding`, `download_arxiv_full_text`, `read_arxiv_chunk`, `cite`, `cite`, `post_question`, `post_finding`, `post_finding`, `read_arxiv_chunk`, `read_arxiv_chunk`, `post_finding`, `post_finding`, `read_arxiv_chunk`, `post_finding`, `cite`, `cite`, `cite`, `post_question`, `read_arxiv_chunk`, `post_finding`, `get_arxiv_metadata`, `get_arxiv_metadata`, `read_arxiv_chunk`, `get_paper_figures`, `search_arxiv_abstracts`, `get_paper_figures`, `read_arxiv_chunk` `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `search_arxiv_abstracts`, `read_arxiv_chunk`, `read_arxiv_chunk`, `search_arxiv_abstr` `get_arxiv_metadata`, `read_arxiv_chunk`, `get_arxiv_metadata`, `read_arxiv_chunk`, `post_finding`, `post_finding`, `post_dead_end`, `post_question`, `post_finding`, `post_finding`, `post_question`, `post_dead_end`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_question`

Survey — 29 tools on 2 servers, 5 read-only probes, 11 papers found.

The literature agrees that a pure higgsino (an SU(2) doublet with Y=1/2) makes up all of the dark matter as a thermal relic at a mass of roughly 1.0-1.2 TeV. Sommerfeld-enhanced coannihilation calculations, for example 1411.6930, move the exact mass within that window, and 2606.31013 quotes about 1 TeV. Pure-higgsino scattering in direct detection comes from loops and is tiny: the one- and two-loop electroweak calculations (1104.0228 and 1007.2601) put the spin-independent cross section well below current xenon limits, near or inside the neutrino fog. I did not read those full texts, so the exact numbers need checking. The 2024 LZ result (2410.17036, 4.2 tonne-years) and the first XENONnT result (2303.14729) are tightest near 30-40 GeV. At 1 TeV their limits scale up roughly linearly with mass, to about 1e-46 to 1e-45 cm^2 (my estimate, not read from their tables). A thermal pure higgsino is therefore not excluded by elastic scattering. Direct detection can see it only if mixing with gauginos raises the Higgs coupling and the mass splittings (2606.31013), or if the splitting between the two neutral states is small enough (roughly up to a few hundred keV) that tree-level inelastic Z-exchange turns on.

That inelastic case is already strongly constrained, and 2609.09830 recomputes it at one loop for LZ. Planck 2018 (1807.06209) sets the target abundance at Omega_c h^2 = 0.120 +/- 0.001.

Quantities worth checking: Omega h^2 against higgsino mass at about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, with and without Sommerfeld enhancement; find the mass where Omega h^2 = 0.120; Charged-neutral splitting (about 340 MeV from electroweak loops in the pure limit) and neutral-neutral splitting delta_m0 as functions of the gaugino masses M1 and M2; Elastic spin-independent cross section of the pure higgsino at 1.1 TeV compared with the LZ 2024 and XENONnT limits at the same mass, and with the xenon neutrino fog; Inelastic Z-exchange threshold: the largest delta_m0 at which xenon at about 1 TeV can still produce a recoil (scale set by the kinematic limit, about mu v^2 / 2), and the delta_m0 values LZ excludes; Rescaling of signal rate by (Omega h^2 / 0.12) for higgsinos lighter than the thermal mass that make up only part of the dark matter; Size of the tree-level Higgs coupling as a function of M1, M2 and tan(beta), to map where mixing makes higgsinos visible to LZ.

Plan — 3 lines of investigation:

1. Establish the pure-higgsino thermal relic density as a function of mass (about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV), with and without Sommerfeld enhancement, and find the mass at which Omega h^2 equals the Planck 2018 value of 0.120 +/- 0.001. — This is the core of the task. The catalog has no inelastic-DM or relic-density compute tool and no coding-task tool, so the Omega h^2(m) curve has to come from published calculations. Each value must be read from text, tables or figures, never estimated.

2. Place the pure-higgsino elastic spin-independent cross section at about 1.0-1.2 TeV against the LZ 2024 (2410.17036) and XENONnT (2303.14729) limits at the same mass, and against the xenon neutrino fog. — This answers the question of where the higgsino sits relative to current limits. The survey's 1e-46 to 1e-45 cm^2 limit at 1 TeV is an unverified estimate and must be replaced with read values.

3. Assess the routes by which a higgsino becomes visible to direct detection: the mass splittings and the inelastic Z-exchange threshold, LZ inelastic constraints (2609.09830), mixing-induced Higgs couplings in the (mu, M1/M2, tan beta) plane (2606.31013), and the (Omega h^2/0.12) rescaling for sub-thermal higgsinos. — Elastic scattering leaves the pure higgsino invisible, so whether it is visible depends on the splittings set by gaugino masses. This direction goes past the base question and is a robustness check on any 'not excluded' conclusion. Critic's view: revise — Step 1 is not enough: the gap has to show up in the final answer. No relic-density or inelastic-DM compute tool exists, list_input_files is empty, and run_test_job only draws random samples. The task asked us to compute relic density with tools, and we cannot. The report must say plainly that every Omega h^2 value is taken from the literature. Do not use run_test_job to produce anything that looks like a computed relic curve.; 1411.6930 is a pMSSM scan. It may not give a pure-higgsino Omega h^2(m) curve at 0.9-1.3 TeV with and without Sommerfeld enhancement. Before relying on it, check that such a figure or table exists. If it does not, use the one-parameter thermal mass plus the m^2 scaling, and label every resulting row as derived, not read. Also note that the higgsino Sommerfeld effect is small, only a few percent shift in mass. Bound-state formation (recent 2021-2026 work) can matter as much as Sommerfeld, so the search should target it explicitly.; The budget is tight. Fifteen calls do not cover three or four full-text downloads plus chunk reads, figures and cite. Drop the full read of 1008.2905, which is a method paper and not needed for numbers. Get the Planck Omega_c h^2 = 0.120 +/- 0.001 quote with one targeted chunk read of 1807.06209. Limit independent cross-checks to one paper..

Wave 1 — now exploring, in parallel: Establish the pure-higgsino thermal relic density as a

function of mass (about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV), with and without Sommerfeld enhancement, and find the mass at which Omega h^2 equals the Planck 2018 value of 0.120 +/- 0.001.; Place the pure-higgsino elastic spin-independent cross section at about 1.0-1.2 TeV against the LZ 2024 (2410.17036) and XENONnT (2303.14729) limits at the same mass, and against the xenon neutrino fog.; Assess the routes by which a higgsino becomes visible to direct detection: the mass splittings and the inelastic Z-exchange threshold, LZ inelastic constraints (2609.09830), mixing-induced Higgs couplings in the (mu, M1/M2, tan beta) plane (2606.31013), and the (Omega h^2/0.12) rescaling for sub-thermal higgsinos..

Finding: Kinematic threshold for inelastic up-scattering on xenon (my arithmetic): m_A = 131 x 0.9315 GeV = 122.0 GeV; mu_A = 108.7 GeV (m=1.0 TeV) and 109.8 GeV (m=1.1 TeV); v_max = 544+232 = 776 km/s, beta^2 = (776/299792)^2 = 6.70e-6; delta_max = mu_A beta^2 /2 = 364 keV (1.0 TeV) and 368 keV (1.1 TeV). The minimum-velocity formula v_min = sqrt(2 delta/mu_A) is given in 2609.09830 Appendix A (Eq. A7).

Finding: 2609.09830 does not quote an LZ-excluded delta_m0 range. Its five benchmarks (|mu| = 0.61-1.40 TeV; |M1|, M2 about 5.8-42 TeV; tan beta 10.9-26.5) have one-loop on-shell splittings delta = 8.9-36.4 keV. Each predicts at most about 1.03 combined NLO events in LZ, and probing them needs 9.4-65.5 tonne-years. All are consistent with the LZ bound. Inelastic events at NLO range from 0.00022 (BP1) to 0.18 (BP4). NLO vertex corrections can exceed 100 percent, and in BP2 the inelastic cross section is about twice the LO elastic one. Splittings of O(10) keV come from partial cancellation of the M_Z^2/M1 and M_Z^2/M2 terms when M1 and M2 have opposite signs.

Finding: 2606.31013: the SI coupling comes from Higgs exchange, f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + tan^2 theta_W/(M1-|mu|)](1 + sgn(mu) sin 2beta), so it is suppressed at large gaugino mass and at small tan beta with mu<0. The scan covers 150<|mu|<1000 GeV, 1.6<tan beta<50, 0.5<M1<10 TeV with M2=2M1. Rates are rescaled by Omega_H h^2/0.12 (Eq. 38, computed with micrOMEGAs 6.3.0). Direct detection is sensitive mainly where higgsino-gaugino mixing produces sizable charged-neutral splittings, and reaches the thermal mass of about 1 TeV. The radiative charged-neutral splitting tends to about 356 MeV for M >> M_Z (Eq. 13).

Finding: Hisano et al. (1104.0228) find the leading-order electroweak-loop SI cross section of pure EW-multiplet DM (including the higgsino, n=2, Y=1/2, evaluated at M=1 TeV) to be O(1e-46 to 1e-48) cm^2, suppressed by an accidental cancellation: twist-2 contributions are positive while Higgs-exchange, gluon and scalar contributions are negative, and they are of similar size. The cross section is nearly independent of mass above about 1 TeV.

Finding: LZ 2024 (2410.17036, 4.2 tonne-years): the strongest SI exclusion is 2.2e-48 cm^2 (90% CL) and the best median sensitivity is 5.1e-48 cm^2, both at 40 GeV/c^2. The abstract quotes no value at 1 TeV.

Finding: No relic-density or inelastic-DM compute tool and no input data tables are available: list_input_files returned an empty file list, and the catalog contains only statistics/histogram, test-job and literature tools. All Omega h^2 values for d-01 therefore come from the literature and were not computed.

Finding: Beneke, Hellmann and Ruiz-Femenia (arXiv:1411.6930, Sec. 3) report Omega h^2 = 0.127 (perturbative) and 0.126 (Sommerfeld, with off-diagonal rates) for a pure-higgsino model compared against the higgsino-like pMSSM point 1627006 at m_chi = 1172.31 GeV. The net Sommerfeld correction for pure higgsinos is almost zero (below 1 percent) because enhancement in the neutral channels cancels suppression in the chi0 chi+- channels. For the pMSSM higgsino-like point (higgsino fraction 0.98, dm+ = 1.8 GeV, dm0 = 9.5 GeV) they find Omega h^2 = 0.108 (perturbative) and 0.100 (Sommerfeld), an 8 percent reduction. The quoted Sommerfeld accuracy is about 5 percent at best.

Completed — Assess the routes by which a higgsino becomes visible to direct

detection: the mass splittings and the inelastic Z-exchange threshold, LZ inelastic constraints (2609.09830), mixing-induced Higgs couplings in the (mu, M1/M2, tan beta) plane (2606.31013), and the (Omega h^2/0.12) rescaling for sub-thermal higgsinos.. Two of the three success criteria are met; the third is only partly met. (1) The inelastic threshold, worked by hand (f-001): m_Xe = 131 x 0.9315 = 122.0 GeV. The reduced mass mu_A is 108.7 GeV at m = 1.0 TeV and 109.8 GeV at 1.1 TeV. With v_max = 544 + 232 = 776 km/s, (v_max/c)^2 = 6.70e-6, so delta_max = mu_A v_max^2/2 is about 364 keV at 1.0 TeV and 368 keV at 1.1 TeV. The minimum-velocity formula v_min = sqrt(2 delta/mu_A) is Eq. A7 of 2609.09830. (2) Partly met: 2609.09830 does not give an LZ-excluded delta_m0 range (f-002). Its five benchmarks have |mu| = 0.61-1.40 TeV, |M1| and M2 of about 5.8-42 TeV, and one-loop on-shell delta = 8.9-36.4 keV. They predict at most about 1.03 combined NLO events in LZ, need 9.4-65.5 t-y of exposure to be probed (Table VII, Fig. 6), and are all consistent with LZ. NLO vertex corrections can exceed 100 percent, and in BP2 the inelastic cross section is about twice the LO elastic one. Splittings of O(10) keV with gaugino masses of O(10) TeV arise only from partial cancellation between the M_Z^2/M1 and M_Z^2/M2 terms when M1 and M2 have opposite signs; generically the paper quotes O(10-100) MeV. My own estimate, using the formula that is not yet sourced (q-05): same-sign M2 = 2 M1 would need M1 of about 1.4e7 GeV to bring delta below 364 keV. Inelastic Z-exchange therefore matters only in tuned corners. (3) Mixing region where elastic SI becomes visible (f-003, 2606.31013): Higgs exchange gives f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + tan^2 theta_W/(M1-|mu|)](1 + sgn(mu) sin 2beta). This is suppressed at large gaugino mass and at small tan beta with mu<0. The scan covers 150<|mu|<1000 GeV, 1.6<tan beta<50 and 0.5<M1<10 TeV with M2 = 2 M1. Direct detection is sensitive where mixing produces larger charged-neutral splittings and reaches the thermal mass of about 1 TeV (Figs. 4 and 5 of that paper; the figures were not extracted). The pure-limit radiative charged-neutral splitting is about 356 MeV (Eq. 13 of 2606.31013), rather than the 340 MeV in the brief. (4) Rescaling: 2606.31013 Eq. 38 uses sigma_eff = (Omega_H h^2/0.12) sigma, computed with micrOMEGAs 6.3.0, and 2609.09830 states that limits weaken proportionally for sub-thermal fractions. Quotes from both papers are cited in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/references.bib. Full texts are at the same run directory as 2609.09830.txt and 2606.31013.txt.

Finding: Chen and Hill (1912.07795) include the O(mW/M) power corrections, perturbative QCD to NNNLO and ChEFT nuclear effects. They find that the pure-higgsino SI cross section has an upper limit below 1e-48 cm^2 in the TeV mass range, below the estimated xenon neutrino floor. This is about 2 orders of magnitude below the LZ 2024 minimum limit of 2.2e-48 cm^2 at 40 GeV. Assuming linear mass scaling of the LZ limit above about 100 GeV (my estimate, not read from the paper), the LZ limit at 1.1 TeV is about 2.2e-48 x (1100/40), roughly 6e-47 cm^2, so the pure higgsino lies at least about 1.8 orders of magnitude below it.

Completed — Place the pure-higgsino elastic spin-independent cross section at about 1.0-1.2 TeV against the LZ 2024 (2410.17036) and XENONnT (2303.14729) limits at the same mass, and against the xenon neutrino fog.. Answer: a thermal pure higgsino at about 1.1 TeV lies about 2 or more orders of magnitude below the LZ 2024 limit and below the xenon neutrino floor. Three of the planned inputs were not read: the XENONnT limit, the LZ limit read from its figure at 1 TeV, and a numeric neutrino-fog value. All are recorded as open question q-06.

(1) Prediction. Hisano et al. 1104.0228 work at leading order in electroweak loops. They find the SI cross section of pure SU(2)-multiplet dark matter is O(1e-46 to 1e-48) cm^2, nearly independent of mass above about 1 TeV. An accidental cancellation drives this: twist-2 terms are positive, while the Higgs-exchange, gluon and Z-box scalar terms are negative and of similar size. Their Fig. 5 shows the higgsino (n=2, Y=1/2) at M = 1 TeV against the Higgs mass; the paper predates the measured Higgs mass. Finding f-004.

The newer and more complete calculation is Chen and Hill 1912.07795, which adds O(mW/M) power corrections, QCD corrections to NNNLO and ChEFT nuclear effects. Its verbatim result is: 'The upper limit for the pure doublet is less than 10^-48 cm^2 in the TeV mass regime ... lower than the estimated Xenon neutrino floor', with 'the pure Higgsino well below the neutrino floor'. The uncertainty is quoted only as this upper bound; no central value plus or minus an error was read. Citation recorded in references.bib (tc-...-64).

(2) LZ 2024 limit (2410.17036). Verbatim: the strongest SI exclusion is 2.2e-48 cm^2 at 90% CL, and the best median sensitivity is 5.1e-48 cm^2, both at 40 GeV. Finding f-005, citation tc-...-65. No 1 TeV value was read from the text or a figure.

(3) Rescaling. Scaling the 40 GeV limit linearly with mass is my estimate, not a read value. It gives about 2.2e-48 x (1100/40), roughly 6e-47 cm^2, at 1.1 TeV. The ratio of limit to predicted upper bound is then at least about 60, i.e. at least 1.8 orders of magnitude. This extrapolation is likely conservative because the limit curve rises from its minimum. Finding f-008, medium confidence.

Not done:

- The XENONnT 2303.14729 1 TeV limit was not read.
- No limit figure was extracted, so there are no figure numbers for the LZ or XENONnT limits.
- No numeric neutrino-fog value at 1 TeV was obtained; the 'below the neutrino floor' statement rests on Chen and Hill's text.
- Projected sensitivities (LZ full exposure, XLZD/DARWIN) were not tabulated.

Also noted: 2607.20588 (Bisal et al., 2026) says next-to-leading-order corrections can push the higgsino cross section below the neutrino floor or make it vanish in parts of MSSM parameter space. This is relevant to q-04.

Files are in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110
1104.0228.txt, 2410.17036.txt, 1912.07795.txt and references.bib.

Finding (preliminary): Hisano et al. (hep-ph/0610249) expect the Sommerfeld effect on the higgsino-like relic abundance to be roughly 10 percent, compared with a 50 percent reduction for the wino. 1411.6930 later found the net pure-higgsino effect is below 1 percent (Omega h^2 = 0.127 perturbative, 0.126 Sommerfeld). Rescaling that single point (taken as m = 1172.31 GeV) with Omega h^2 proportional to m^2 gives the Planck value 0.120 at m of about 1.14 TeV (Sommerfeld: 1144 GeV; perturbative: 1139 GeV). The rescaled values are 0.074, 0.092, 0.111, 0.132 and 0.155 at m = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV. These are m^2 rescalings of one published point, not published values.

Blocked — Establish the pure-higgsino thermal relic density as a function of mass (about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV), with and without Sommerfeld enhancement, and find the mass at which Omega h^2 equals the Planck 2018 value of 0.120 +/- 0.001.. (Tool-call budget used up. No tool can compute relic densities, and 1411.6930 gives only one pure-higgsino mass point, so Omega h^2 at several masses and a second independent thermal mass were not obtained. Next steps: read 1807.06209.txt (already downloaded) and cite the 0.120 +/- 0.001 value, then find a paper with Omega h^2 for higgsinos at several masses, for example later work that includes bound-state formation.) Not finished: only one published pure-higgsino relic-density point was found, so the success criteria (published values at three or more masses, and a thermal mass from two independent papers) are not met.

(1) No relic-density or inelastic-DM compute tool exists here, and list_input_files returned no data files (tc-261016_1-w1-d-01-19, finding f-006). Every value below comes from the literature. Nothing was computed.

(2) arXiv:1411.6930, Sec. 3, gives the one published point: a pure higgsino (all sfermions and heavy Higgses decoupled) has Omega h^2 = 0.127 without and 0.126 with Sommerfeld enhancement (off-diagonal rates included). The net Sommerfeld effect is below 1 percent because the neutral channels are enhanced and the charged ones suppressed. The paper's mass for this model appears to be 1172.31 GeV, the mass of its higgsino-like pMSSM point 1627006; I inferred this from context, the paper does not state it. For that pMSSM point (dm+ = 1.8 GeV, dm0 = 9.5 GeV) the values are 0.108 without and 0.100 with Sommerfeld, an 8 percent drop. The paper quotes about 5 percent accuracy at best and compares to Planck 2013 (0.1187 +/- 0.0017). Its Fig. 7 (registered as Figure 6, figures_1411.6930/fig07.png) shows only the yield ratio Y/Ypert, not Omega h^2 against mass (finding f-007). A verbatim quote is in references.bib (key arxiv14116930).

(3) hep-ph/0610249 expected the Sommerfeld effect for higgsinos to be roughly 10 percent (versus 50 percent for winos). It was not a full calculation (f-009).

(4) Interpolation, not published values: scaling the 1411.6930 point as Omega h^2 proportional to m^2 gives 0.074, 0.092, 0.111, 0.132 and 0.155 at 0.9, 1.0, 1.1, 1.2 and 1.3 TeV. Omega h^2 = 0.120 falls at about 1.14 TeV (1144 GeV with Sommerfeld, 1139 GeV without) (f-009, low confidence).

Still to do: the Planck 2018 text was downloaded but not yet read or quoted; 1008.2905 was not read; no independent recent thermal-mass determination was found.

Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/ 1411.6930.txt, hep-ph_0610249.txt, 1807.06209.txt, figures_1411.6930/fig07.png, references.bib.

Critic, wave 1 — continue: 5 findings stand up to the evidence, 4 questioned ('2606.31013: the SI coupling comes from Higgs exchange, f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + t': None of the three quoted excerpts contains any claimed content. tc-...-35 covers the oblique-parameter chi-square (Eqs. 26-29), tc-...-30 is the title page, and tc-...-33 covers the W/Y parameters (Eqs. 39-40). Missing are the f_q formula, the scan ranges, Eq. 13 (356 MeV), Eq. 38 with micrOMEGAs 6.3.0, and the statement that direct detection reaches about 1 TeV only with mixing. These may well be in the paper, but the cited evidence does not show them.; 'Beneke, Hellmann and Ruiz-Femenia (arXiv:1411.6930, Sec. 3) report Omega h^2 = 0.127 (pert': The excerpts do not show the Omega h^2 values 0.127, 0.126, 0.108 or 0.100, nor dm+ = 1.8 GeV, dm0 = 9.5 GeV or the 5 percent accuracy. The claim states m_chi = 1172.31 GeV as reported, but the d-01 summary says this mass was inferred from context and is not stated in the paper. The figure is also mislabelled: fig07 is registered as Figure 6.; 'Chen and Hill (1912.07795) include the O(mW/M) power corrections, perturbative QCD to NNNL': There are two problems. First, the Chen and Hill verbatim quote (the upper limit below 1e-48 cm^2, below the neutrino floor) is not in any quoted excerpt. The cited tc-...-59 and -58 show only the rate formula and the title, and the citation call tc-...-64 is not listed as evidence. Second, there is an arithmetic error. An upper bound below 1e-48 cm^2 compared with the 2.2e-48 cm^2 minimum is a factor of about 2 or more, not 'about 2 orders of magnitude'. Only the comparison at 1.1 TeV (about 6e-47 cm^2) supports a margin of 1.8 orders or more.; '2609.09830 does not quote an LZ-excluded delta_m0 range. Its five benchmarks (|mu| = 0.61-': The visible part of Table VII supports only BP1 elastic NLO = 1.028 events, which is consistent with 'at most about 1.03'. The other quantities are not in the shown excerpts: the splitting range 8.9-36.4 keV, the exposures of 9.4-65.5 t-y, the inelastic NLO range, and the BP2 statement. The visible LO numbers for BP2 (inelastic 0.0247, elastic 0.0436) show inelastic below elastic. The 'twice the LO elastic' claim therefore needs the NLO inelastic value quoted explicitly.).

Issues: The core quantity is still unmet. There is no published Omega h^2(m) curve at multiple masses and no second, independent determination of the thermal mass. The m^2 rescaling in f-009 depends on an inferred mass of 1172.31 GeV. If that inference is wrong, every rescaled value and the 1.14 TeV thermal mass shift with it.; The Planck 2018 value of 0.120 +/- 0.001 is used throughout but has never been quoted from 1807.06209.txt, although the file is already downloaded. 1411.6930 normalised to Planck 2013 (0.1187), so the f-009 rescaling should state which target it uses.; The arithmetic checks out: 0.126 x (m/1172.31)^2 gives 0.074, 0.092, 0.111, 0.132 and 0.155, and the thermal masses of 1144 and 1139 GeV are correct. However, the hep-ph/0610249 '10 percent' statement is not visible in the tc-...-68 excerpt..

Next steps after wave 1 — This is the final wave, with 19 tool calls across at most 3 directions. d-01 is closed and replaced by a narrower child that works mostly from files already downloaded: quote Planck 2018, confirm the 1411.6930 mass, and add one independent thermal-mass source that includes bound-state formation. The limits direction carries the headline answer, so it gets priority 1. It reads LZ and XENONnT at 1 TeV from their figures, cites Chen and Hill verbatim, adds fog and XLZD numbers, and fixes the f-008 arithmetic. The third direction is a lower-priority verification pass on the contested f-002 and f-003 evidence and the unsourced delta_m0 formula (q-02, q-05). No compute tool for relic density exists (f-006, q-01), so the final report must state that all Omega h^2 values come from the literature or are labelled derived rescalings. The task does not ask for facility runs, so none is scheduled. Each direction should use about 6 calls and reuse already-downloaded texts wherever possible.

New lines of investigation: Pin down the pure-higgsino thermal mass from at least two independent published calculations, including recent work with bound-state formation, and quote the Planck 2018 Omega_c h^2 verbatim. Confirm which mass the 1411.6930 pure-higgsino values (0.126/0.127) refer to.; Replace the linear 40 GeV extrapolation with direct reads of the LZ 2024 and XENONnT SI limits at about 1 TeV, a numeric xenon neutrino-fog value at about 1 TeV, and one next-generation projection (XLZD/DARWIN). Correct the f-008 arithmetic and quote Chen and Hill verbatim.; Verify 2606.31013 and 2609.09830 with metadata, and trace the contested f-002 and f-003 numbers to quoted excerpts: Eq. 13 (356 MeV), the f_q formula, Eq. 38, the Table VII NLO rows, and the tree-level delta_m0 formula (Eq. 9 / Ref. [28])..

Wave 2 — now exploring, in parallel: Pin down the pure-higgsino thermal mass from at least two independent published calculations, including recent work with bound-state formation, and quote the Planck 2018 Omega_c h^2 verbatim. Confirm which mass the 1411.6930 pure-higgsino values (0.126/0.127) refer to.; Replace the linear 40 GeV extrapolation with direct reads of the LZ 2024 and XENONnT SI limits at about 1 TeV, a numeric xenon neutrino-fog value at about 1 TeV, and one next-generation projection (XLZD/DARWIN). Correct the f-008 arithmetic and quote Chen and Hill verbatim.; Verify 2606.31013 and 2609.09830 with metadata, and trace the contested f-002 and f-003 numbers to quoted excerpts: Eq. 13 (356 MeV), the f_q formula, Eq. 38, the Table VII NLO rows, and the tree-level delta_m0 formula (Eq. 9 / Ref. [28])..

Finding: Chen and Hill (1912.07795), verbatim from the Summary: "The upper limit for the pure doublet is less than 10-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than the estimated Xenon neutrino ?oor [26]." Verbatim from Sec. 4: "In the pure Higgsino limit, the cross section upper limit remains at or below 10 -48 cm2 for masses above a few hundred GeV." Abstract: "The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range." Sec. 4 also says that away from the pure limit, with c_H/g2^2 of about 0.1, the TeV cross section is within reach of next-generation detectors, and with c_H/g2^2 of about 0.01 it is close to the neutrino floor.

Finding: Correction to f-008: an upper bound below 1e-48 cm^2 compared with the LZ minimum of 2.2e-48 cm^2 (at 40 GeV) is a factor of more than 2.2 (more than 0.34 decades), not 2 orders of magnitude. The comparison that actually matters is at the same mass. At 1.1

TeV the linearly rescaled LZ limit of about 6e-47 cm^2, divided by the bound of 1e-48 cm^2, gives a ratio of at least about 60 (at least about 1.8 decades). That ratio is still an estimate: the LZ and XENONnT curves at 1 TeV were not read.

Finding: Planck 2018 (arXiv:1807.06209, abstract, page 1) states verbatim: "A combined analysis gives dark matter density Ωch2 = 0.120± 0.001" (68% confidence region), quoted for base-LambdaCDM from polarization, temperature and lensing combined.

Finding: Bottaro et al., "The last Complex WIMPs standing" (arXiv:2205.04486, 2022), is the independent recent calculation of thermal masses for complex electroweak n-plets, which includes the Y=1/2 doublet (the higgsino). Its companion paper arXiv:2107.09688 states that the method includes "Sommerfeld enhancement and bound states formation at leading order", and 2205.04486 says the doublet with hypercharge 1/2 is the notable exception that future direct detection cannot fully probe. The numeric doublet thermal mass was not read in this direction.

Blocked — Replace the linear 40 GeV extrapolation with direct reads of the LZ 2024 and XENONnT SI limits at about 1 TeV, a numeric xenon neutrino-fog value at about 1 TeV, and one next-generation projection (XLZD/DARWIN). Correct the f-008 arithmetic and quote Chen and Hill verbatim.. (The figure-extraction tool returned none of the SI limit plots from 2410.17036 or 2303.14729, and the tool budget ran out before the rest of the LZ text, the XENONnT text, HEPData or the XLZD Design Book (2410.17137) could be read.) Two things were established: the Chen and Hill quotes and the corrected f-008 arithmetic. The 1 TeV limits, the neutrino-fog value and the XLZD projection were not obtained. (1) Chen and Hill (arXiv:1912.07795), verbatim from the Summary: 'The upper limit for the pure doublet is less than 10^-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than the estimated Xenon neutrino floor [26].' From Sec. 4: 'In the pure Higgsino limit, the cross section upper limit remains at or below 10^-48 cm2 for masses above a few hundred GeV.' Sec. 4 also says that with gaugino admixture c_H/g2^2 of about 0.1 the TeV cross section is within reach of next-generation detectors, and with about 0.01 it is close to the neutrino floor. Finding f-010. These quotes are in the board finding only; the cite tool was not called in this direction, so they were not added to references.bib. (2) Correction to f-008: an upper bound below 1e-48 cm^2 compared with the LZ minimum of 2.2e-48 cm^2 at 40 GeV is a factor of more than 2.2 (more than 0.34 decades), not 2 orders of magnitude. Comparing at the same mass, the linearly rescaled LZ limit at 1.1 TeV (about 6e-47 cm^2, an estimate) is at least about 60 times the bound (at least about 1.8 decades). Finding f-011. (3) Not obtained: get_paper_figures did not return the limit plots (dead end x-01). For 2410.17036 it returned only Fig. 1 and Fig. S4, registered as Figures 10 and 11. For 2303.14729 it returned only calibration and spatial-distribution plots, registered as Figures 12-14. The LZ text chunk at characters 20000-28000 contained no 1 TeV value. The XLZD Design Book (arXiv:2410.17137) was identified but not read, so there are no fog or projection numbers yet (question q-07). The LZ/prediction, XENONnT/prediction and fog/prediction ratios at 1.1 TeV therefore remain estimates or are missing. Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/: figures_2410.17036/, figures_2303.14729/, 1912.07795.txt, 2410.17036.txt.

Blocked — Pin down the pure-higgsino thermal mass from at least two independent published calculations, including recent work with bound-state formation, and quote the Planck 2018 Omega_c h^2 verbatim. Confirm which mass the 1411.6930 pure-higgsino values (0.126/0.127) refer to.. (The tool budget ran out before the 2205.04486 full text (thermal-mass table) and 1411.6930 Sec. 3 could be read. A follow-up should download 2205.04486 and read its doublet thermal mass, and search 1411.6930.txt for the pure-higgsino mass statement.) One of the three success criteria is met. (1) Met: the Planck 2018 value is quoted verbatim from the abstract on page 1 of arXiv:1807.06209: 'A combined analysis gives dark matter density Ωch2 = 0.120± 0.001' (68% CL). It is at roughly charac-

ters 6500-7000 of /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt (finding f-012, tc-261016_1-w2-d-04-82). It has not yet been added to references.bib with cite(). (2) Not met: a second independent calculation was identified but its number was not read. Bottaro et al., arXiv:2205.04486 ('The last Complex WIMPs standing', 2022), computes thermal masses for complex electroweak n-plets, including the Y=1/2 doublet (the higgsino). Its companion paper, arXiv:2107.09688, describes the method as including Sommerfeld enhancement and bound-state formation at leading order (finding f-013, medium confidence, abstracts only). Its doublet thermal mass was not read, so the spread against the 1411.6930-derived value of about 1.14 TeV (f-009) cannot be stated. For now, only one thermal-mass determination is available, and that one is a rescaled value. (3) Not met: the 1411.6930 mass attribution was not re-read in this direction. The 1172.31 GeV mass behind the pure-higgsino values 0.127/0.126 is still inferred, not stated verbatim (f-007 stays contested). The f-009 rescaling to 0.120 still gives about 1144 GeV (Sommerfeld) and 1139 GeV (perturbative). These numbers are derived, not published, and they depend on the inferred mass. One dead end was recorded: the abstract search for recent thermal-mass papers returned only unrelated LEP notes (x-02). Open question q-08 asks for the 2205.04486 doublet mass and its uncertainty.

Finding: Metadata confirms both IDs: arXiv:2606.31013 is 'Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection' by Hamaguchi, Nagata and Osaki (2026), and arXiv:2609.09830 is 'Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter' by Chatterjee, Das, Pasha, Pukhov and Puri (2026). Both abstracts match how the survey uses them. This resolves q-02.

Finding: The full text of 2606.31013 supports the f-003 numbers. Eq. 13 reads Delta m+|rad ~ (1/2) alpha2 MZ sin^2 thetaW (1 - 3MZ/(2 pi M_chi+)), with the statement 'For M_chi+ >> MZ, this mass splitting asymptotically approaches Delta m+|rad ~ 356 MeV'. Eq. 33 reads f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + tan^2 thetaW/(M1-|mu|)](1 + sgn(mu) sin 2beta) and is valid for MZ << M1,2-|mu|; the exact form is Eq. 32. The paper says the coupling is suppressed at large gaugino mass and 'for small tan beta and negative mu, due to an accidental cancellation'. Eq. 38 is sigma_SI/SD,eff = (Omega_H h^2/0.12) sigma_SI/SD, computed with micrOMEGAs v6.3.0. The scan (Eq. 37) covers 150 GeV<|mu|<1000 GeV, 1.6<tan beta<50 and 0.5 TeV<M1<10 TeV, with M2=2M1 and M1>|mu|. Note: the f-003 label 'Eq. 13' is correct; the f_q formula is Eq. 33, not unnumbered.

Finding: The full text of 2609.09830 supports the f-002 numbers. Table III gives one-loop on-shell delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5, with mu = -739.63, -606.64, -814.26, 1282.11 and 1400.0 GeV, M1 = -7830, -6292, -5838, 11354 and 12875 GeV, M2 = 17055, 42000, 30925, 12627 and 12050 GeV, and tan beta = 10.88-26.46. Table VII, NLO rows for BP1-BP5: elastic events 1.0280, 0.07729, 0.1176, 0.4127, 0.5074; inelastic events 0.00022, 0.07039, 0.0518, 0.1806, 0.1661; combined 1.02822, 0.14768, 0.1694, 0.5933, 0.6735; required exposure 9.41, 65.50, 57.10, 16.30, 14.36 t-y (LZ exposure 4.2 t-y; null result taken as N_E < ln 10, about 2.3). Table VI for BP2 gives sigma_inel NLO = 0.507e-12 pb against sigma_el LO = 0.284e-12 pb (ratio 1.785). The paper describes this as 'approximately a factor of two larger than the elastic LO one'. The text says all BPs are consistent with LZ. Note: the BP1 scan sign pattern has M1<0 and M2>0 for mu=±0.5 or ±1 TeV, with -21 TeV <= M1 <= -1 TeV and 1 TeV <= M2 <= 21 TeV (Eq. 25), yet BP4 and BP5 have M1 and M2 both positive.

Finding: 2609.09830 does not write out the tree-level delta_m0 formula. It says only: 'as can be seen, for example, from Eq. (9) of Ref. [28], the corrections proportional to M_Z^2/M1 and M_Z^2/M2 depend on the relative signs of M1 and M2. In the parameter regions relevant for our analysis, these contributions can have opposite signs and partially cancel.' 2606.31013 also says only that the splittings are 'suppressed by M_Z^2/M_1,2'. The explicit form delta_m0 ~ m_Z^2 s_W^2/M1 + m_Z^2 c_W^2/M2 therefore remains unsourced (q-05 stays open), and

so does the estimate that same-sign M2=2M1 needs M1 of about 1.4e7 GeV. Counter-evidence to the 'tuned corners only' conclusion: BP4 and BP5 reach delta of about 10.6-10.8 keV with M1 and M2 both positive (about 11-13 TeV) and mu > 0. The O(10) keV splittings there come from on-shell one-loop corrections combined with relative signs between mu and the gauginos, not only from opposite-sign M1 and M2.

Completed — Verify 2606.31013 and 2609.09830 with metadata, and trace the contested f-002 and f-003 numbers to quoted excerpts: Eq. 13 (356 MeV), the f_q formula, Eq. 38, the Table VII NLO rows, and the tree-level delta_m0 formula (Eq. 9 / Ref. [28]).. Both papers are real, and every contested f-002/f-003 number except the tree-level delta_m0 formula is now traced to a verbatim excerpt. That formula is still unsourced, so the M1 needed for delta < 364 keV was not re-derived. No cite() calls were made: the quotes are recorded on the board as f-015 and f-016, not in references.bib.

(1) IDs confirmed by metadata (f-014, closes q-02). 2606.31013 is Hamaguchi, Nagata and Osaki (2026), 'Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection'. 2609.09830 is Chatterjee, Das, Pasha, Pukhov and Puri (2026), 'Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter'.

(2) 2606.31013 (f-015):

- Eq. 13: Delta m+ ~ (1/2) alpha2 MZ sin^2 thetaW (1 - 3MZ/(2 pi M)), which tends to about 356 MeV for M >> MZ.
- The approximate f_q formula is Eq. 33 (the exact form is Eq. 32). It is suppressed at large gaugino mass, and at small tan beta with mu < 0 through an accidental cancellation.
- Eq. 38: sigma_eff = (Omega_H h^2/0.12) sigma, computed with micrOMEGAs v6.3.0.
- Eq. 37 scan: 150 GeV < |mu| < 1000 GeV, 1.6 < tan beta < 50, 0.5 TeV < M1 < 10 TeV, with M2 = 2 M1.

(3) 2609.09830 (f-016):

- Table III: one-loop on-shell delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5.
- Table VII, NLO events for BP1-BP5:
- inelastic: 0.00022, 0.07039, 0.0518, 0.1806, 0.1661
- elastic: 1.0280, 0.07729, 0.1176, 0.4127, 0.5074
- combined: 1.02822, 0.14768, 0.1694, 0.5933, 0.6735
- exposure needed to probe each point: 9.41, 65.50, 57.10, 16.30, 14.36 t-y (LZ has 4.2 t-y; the null-result limit used is N < ln 10, about 2.3).
- Table VI, BP2: NLO inelastic cross section 0.507e-12 pb against LO elastic 0.284e-12 pb, a ratio of 1.785.

(4) delta_m0 formula (f-017). 2609.09830 only points to Eq. (9) of its Ref. [28]. It says the M_Z^2/M1 and M_Z^2/M2 terms depend on the relative signs of M1 and M2 and can partially cancel, but it does not write out the formula. So q-05 stays open, and so does the earlier estimate that same-sign M2 = 2 M1 needs M1 of about 1.4e7 GeV for delta < 364 keV.

(5) Correction to f-002 and the 'tuned corners only' conclusion: BP4 and BP5 reach about 10.6-10.8 keV with M1 and M2 both positive (about 11-13 TeV) and mu > 0. Opposite-sign gauginos are therefore not the only route to O(10) keV splittings; on-shell one-loop corrections and relative signs between mu and the gauginos also contribute.

Open: q-09, identify Ref. [28] of 2609.09830 and quote its Eq. (9).

Full texts: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_1104 and 2609.09830.txt.

Critic, wave 2 — continue: 7 findings stand up to the evidence, 1 questioned ('Chen and Hill (1912.07795), verbatim from the Summary: "The upper limit for the pure doubl': Neither excerpt contains the quoted text. tc-...-76 (chars 0-3000) stops partway through the abstract, before the quoted abstract sentence. tc-...-81 (chars 22000-29000) covers the uncertainty methodology only. The Summary sentence ('less than 10^-48 cm2 ... lower than the estimated Xenon neutrino floor') and the Sec. 4 sentences, including the c_H/g2^2 = 0.1 and 0.01 statements, are not shown. In a 37078-character document the Summary likely sits after char 29000, so it is probably outside both retrieved ranges. The quotes are plausibly correct and were earlier cited in d-02 (tc-...-64), but that call is not attached here. They need to be re-read with an excerpt that actually shows the text.).

Issues: The arithmetic is correct: 2.2e-48 / 1e-48 = 2.2, which is 0.34 decades, and 2.2e-48 x 1100/40 = 6.05e-47, giving a ratio of at least about 60 (about 1.78 decades). The key number is still a linear extrapolation from the 40 GeV minimum, not a read value. The 2.2e-48 input comes from f-005 and is not in this finding's excerpts. The headline 'where does the higgsino sit' ratio therefore still lacks a measured limit at 1 TeV.; The excerpt from tc-...-82 is truncated before the quoted sentence. The retrieved range (chars 5500-10000) does cover the stated location (about 6500-7000), and the value is standard, so it is accepted. It is still not in references.bib; cite() should be called.; None of the Table III, VI or VII numbers appear in the displayed excerpts. Only surrounding prose is shown, although the retrieved ranges (chars 27000-50000) plausibly contain the tables. This was the original reason f-002 was contested, so the trace is only partly verbatim. f-016 is accepted on the strength of the range coverage and the consistent BP3-BP5 prose. The numbers should still be recorded with cite() and table excerpts..

Stopping after wave 2: budget reached (2/2 waves used). Writing the report from the board.

Research report:

The thermal pure higgsino (about 1.1 TeV) against Planck 2018, LZ 2024 and XENONnT: a literature survey

Abstract

The task was to compute the pure-higgsino relic density at several masses with inelastic-DM tools, and to place the 1.0–1.2 TeV thermal higgsino against LZ, XENONnT and the Planck abundance. No relic-density or inelastic-DM compute tool was available in this environment. No input data were available either. **Every Omega h^2 value below therefore comes from the literature, or is a labelled rescaling of a literature value; none was computed.**

The main findings are these:

- **Relic density.** The one published pure-higgsino point found is Omega h^2 = 0.127 (perturbative) and 0.126 (Sommerfeld) in arXiv:1411.6930. The mass of about 1172 GeV attached to it is inferred from context, not stated. Rescaling that point as m^2 to the verbatim Planck 2018 value, Omega_c h^2 = 0.120 ± 0.001, gives a thermal mass of about 1.14 TeV.

- **Elastic scattering.** The most complete elastic calculation (arXiv:1912.07795) bounds the pure-higgsino spin-independent (SI) cross section below 1e-48 cm^2 at TeV masses, below the xenon neutrino floor. LZ 2024 is tightest at 40 GeV, with 2.2e-48 cm^2. A linear extrapolation to 1.1 TeV, which is an estimate and not a read value, puts the LZ limit at about 6e-47 cm^2. That is at least about 60 times above the higgsino bound.
- **Inelastic scattering.** The kinematic threshold for up-scattering on xenon is about 364–368 keV at 1.0–1.1 TeV. Recent one-loop benchmarks with splittings of 9–36 keV predict at most about one LZ event and are all allowed.

Net conclusion: a thermal pure higgsino is not excluded by current direct detection. It becomes visible only through gaugino mixing or very small neutral splittings.

Background

A pure higgsino is an SU(2) doublet with Y = 1/2. The literature places its thermal-relic mass at roughly 1.0–1.2 TeV, with the exact value set by the Sommerfeld and coannihilation treatment:

- Beneke, Hellmann and Ruiz-Femenia, arXiv:1411.6930, give a Sommerfeld-enhanced coannihilation calculation.
- arXiv:2606.31013 quotes about 1 TeV.
- An earlier estimate, hep-ph/0610249, expected a Sommerfeld effect of about 10% for higgsinos, against about 50% for winos.
- A more recent independent framework includes bound-state formation as well as Sommerfeld enhancement (Bottaro et al., arXiv:2107.09688 and arXiv:2205.04486).
- The target abundance is from Planck 2018 (arXiv:1807.06209).

Pure-higgsino elastic scattering arises only at loop level and is suppressed by an accidental cancellation:

- Hisano et al. give the leading-order result in arXiv:1104.0228, with gluon contributions in arXiv:1007.2601.
- Chen and Hill, arXiv:1912.07795, refine it with power corrections, QCD corrections to NNNLO and chiral-EFT nuclear effects.
- arXiv:2607.20588 (2026) argues that NLO corrections can push the cross section lower, or make it vanish, in parts of MSSM parameter space.

The experimental references are:

- LZ, arXiv:2410.17036, with 4.2 tonne-years of exposure.
- XENONnT, arXiv:2303.14729.
- The ICRC 2025 review, arXiv:2510.17473, which notes that xenon experiments are entering the solar-neutrino-limited regime.

Two 2026 papers cover the visibility routes:

- **Mixing (arXiv:2606.31013).** Hamaguchi, Nagata and Osaki show that gaugino mixing raises the Higgs coupling.
- **Inelastic scattering (arXiv:2609.09830).** Chatterjee et al. compute inelastic higgsino scattering at one loop for LZ.

Both IDs were confirmed by metadata lookup.

Results

Planck target and the absence of a compute tool Planck 2018 states verbatim, on page 1 of the abstract: "A combined analysis gives dark matter density Ωch2 = 0.120± 0.001" (68% CL, base ΛCDM). This value was read from the downloaded text but not added to the bibliography file.

A file listing returned an empty input set. The tool catalog contains only statistics/histogram, test-job and literature tools. The task's request to "compute the relic density" therefore could not be carried out.

Relic density against mass arXiv:1411.6930 (Sec. 3) reports the following:

- **Pure-higgsino model** (sfermions and heavy Higgses decoupled): Omega h^2 = 0.127 without and 0.126 with Sommerfeld enhancement, including off-diagonal rates. The net Sommerfeld effect is below 1%, because enhancement in the neutral channels cancels suppression in the neutral–charged channels.
- **Higgsino-like pMSSM point 1627006** (higgsino fraction 0.98, Δm+ = 1.8 GeV, Δm0 = 9.5 GeV): Omega h^2 = 0.108 perturbative and 0.100 with Sommerfeld, an 8% reduction.
- **Normalisation and accuracy.** The paper quotes about 5% Sommerfeld accuracy at best and normalises to Planck 2013 (0.1187 ± 0.0017).

The pure-higgsino mass of 1172.31 GeV is the mass of the pMSSM comparison point. Its attribution to the pure-higgsino numbers was inferred from context and is not stated in the paper. The critic flagged this, and it remains unconfirmed.

[[FIGURE 6]]

Figure 6 reproduces Fig. 7 of arXiv:1411.6930. It shows the yield ratio Y/Y_pert, Sommerfeld over perturbative, for the pMSSM higgsino-like point with and without off-diagonal rates, and for the pure-higgsino model. It shows the size of the Sommerfeld correction, not Omega h^2 against mass. No published Omega h^2(m) curve was found.

The table below is an m^2 rescaling of the single point (0.126 × (m/1172.31 GeV)^2). It is an interpolation, not published values, and it depends on the inferred mass.

m (TeV)	0.9	1.0	1.1	1.2	1.3
Omega h^2 (rescaled)	0.074	0.092	0.111	0.132	0.155

Omega h^2 = 0.120 falls at about 1144 GeV (Sommerfeld) or 1139 GeV (perturbative), that is about 1.14 TeV.

A second, independent thermal mass was identified but not read: the doublet entry of

arXiv:2205.04486. That paper also states that the Y = 1/2 doublet is the notable case that future direct detection cannot fully probe.

Elastic SI cross section against LZ and XENONnT **Prediction.** Hisano et al. (arXiv:1104.0228) find an SI cross section of O(1e-46–1e-48) cm^2 at leading order for pure electroweak multiplets, nearly mass-independent above about 1 TeV. The positive twist-2 terms cancel against the negative Higgs, gluon and scalar terms. The critic noted that the numeric range was not visible in the excerpts read. The calculation scans the Higgs mass and predates its measurement.

Chen and Hill (arXiv:1912.07795) state verbatim: "The upper limit for the pure doublet is less than 10^-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than the estimated Xenon neutrino floor [26]." Sec. 4 adds that the bound "remains at or below 10^-48 cm2 for masses above a few hundred GeV". The same section says that with gaugino admixture c_H/g2^2 of about 0.1 the TeV rate comes within next-generation reach, and with about 0.01 it lies near the floor. These quotes were recorded, but the critic did not treat them as fully verified, and only the earlier citation entry is in the bibliography. Only an upper bound was obtained, not a central value with an uncertainty.

Limits. LZ 2024 gives its strongest SI exclusion as 2.2e-48 cm^2 (90% CL), with a best median sensitivity of 5.1e-48 cm^2, both at 40 GeV. No 1 TeV value was read.

Comparison. Comparing the 1e-48 cm^2 bound with the 40 GeV minimum gives a factor of only more than 2.2 (more than 0.34 decades). An earlier "two orders of magnitude" statement was wrong on this point and has been corrected.

At equal mass, scaling the LZ limit linearly gives 2.2e-48 × 1100/40 ≈ 6e-47 cm^2 at 1.1 TeV. That is a ratio of at least about 60 (at least about 1.8 decades) above the higgsino bound. This is an extrapolation. The XENONnT 1 TeV limit, the LZ 1 TeV curve value and a numeric neutrino-fog value were not obtained.

Inelastic threshold and LZ inelastic sensitivity **Threshold.** The threshold was worked out by hand:

- m_Xe = 131 × 0.9315 GeV = 122.0 GeV.
- μ_A = 108.7 GeV at 1.0 TeV and 109.8 GeV at 1.1 TeV.
- v_max = 544 + 232 = 776 km/s, so β^2 = 6.70e-6.
- δ_max = μ_A β^2/2 gives 364 keV at 1.0 TeV and 368 keV at 1.1 TeV.

The relation v_min = sqrt($2\delta/\mu$_A) is Eq. A7 of arXiv:2609.09830. The result depends on the escape speed and Earth velocity assumed.

Benchmarks. arXiv:2609.09830 does not give an LZ-excluded δ_m0 range. Its five benchmarks are:

- Table III: μ = -739.63, -606.64, -814.26, 1282.11 and 1400.0 GeV; M1 = -7830, -6292, -5838, 11354 and 12875 GeV; M2 = 17055, 42000, 30925, 12627 and 12050 GeV; tan β = 10.88–26.46.
- One-loop on-shell δ = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1–BP5.

Table VII gives the NLO LZ event counts:

	BP1	BP2	BP3	BP4	BP5
Elastic	1.0280	0.07729	0.1176	0.4127	0.5074
Inelastic	0.00022	0.07039	0.0518	0.1806	0.1661
Combined	1.02822	0.14768	0.1694	0.5933	0.6735
Exposure needed (t-y)	9.41	65.50	57.10	16.30	14.36

For comparison, LZ has 4.2 t-y, and the null-result criterion used is N $<$ ln 10 $\approx$ 2.3. All benchmarks are consistent with LZ.

NLO vertex corrections can exceed 100%. For BP2, the NLO inelastic cross section is 0.507e-12 pb, against an LO elastic cross section of 0.284e-12 pb (ratio 1.785).

The critic noted that the table rows were not visible in the displayed excerpts, although the ranges read cover them. They are accepted on that basis.

Splittings and mixing **Charged–neutral splitting.** Eq. 13 of arXiv:2606.31013 gives the pure-limit radiative splitting as Δm+ $\approx$ (1/2) α2 MZ sin^2θW (1 - 3MZ/(2π M)). This tends to about 356 MeV for M ? MZ, rather than the 340 MeV in the brief.

Higgs coupling. Eq. 33 (approximate form of Eq. 32) gives f_q $\approx$ -(g^2/8M_h^2)[1/(M2-|μ|) + tan^2θW/(M1-|μ|)](1 + sgn(μ) sin 2β). This is suppressed at large gaugino mass, and at small tan β with $\mu < 0$ through an accidental cancellation.

Scan. The scan in Eq. 37 covers 150 $<$ |μ| $<$ 1000 GeV, 1.6 $<$ tan β $<$ 50 and 0.5 $<$ M1 $<$ 10 TeV, with M2 = 2 M1. Rates are rescaled as σ_eff = (Omega_H h^2/0.12) σ (Eq. 38, micrOMEGAs v6.3.0). arXiv:2609.09830 likewise weakens its limits proportionally for sub-thermal fractions.

Direct detection is sensitive mainly where mixing produces sizable charged–neutral splittings, and in that region it reaches the about 1 TeV thermal mass. The reach figures of that paper were not extracted.

Neutral splitting δ_m0. The tree-level formula was not sourced. arXiv:2609.09830 refers only to "Eq. (9) of Ref. [28]" and states that the M_Z^2/M1 and M_Z^2/M2 terms can partially cancel. An earlier estimate, that same-sign M2 = 2 M1 needs M1 $\approx$ 1.4e7 GeV for $\delta <$ 364 keV, rests on that unsourced formula and should be disregarded. The critic noted that a same-sign tree-level splitting at about 12 TeV would be only about 0.5–0.7 MeV.

BP4 and BP5 reach about 10.6–10.8 keV with M1 and M2 both positive (about 11–13 TeV) and $\mu > 0$. So the earlier claim that only opposite-sign gauginos give O(10) keV splittings was revised. On-shell one-loop corrections and the relative signs between μ and the gauginos also contribute.

Discussion

The survey supports the brief's qualitative picture.

- **Thermal mass.** A thermal pure higgsino sits near 1.1 TeV. The single rescaled point gives about 1.14 TeV, inside the 1.0–1.2 TeV window. The Sommerfeld effect is negligible in the pure limit (below 1%), smaller than the earlier 10% expectation of hep-ph/0610249.
- **Elastic scattering.** The elastic SI rate is bounded below 1e-48 cm^2. By the extrapolation above, it lies roughly two decades below the current LZ reach at 1.1 TeV and below the

xenon floor. Pure-higgsino elastic scattering is therefore not a viable discovery channel for current or, per arXiv:2205.04486 and arXiv:1912.07795, future xenon detectors.

- **Visibility routes.** The higgsino becomes visible through gaugino mixing, which raises f_q and the splittings (arXiv:2606.31013). It can also become visible through neutral splittings well below the about 365 keV kinematic limit. Even at 9–36 keV, however, the one-loop benchmarks of arXiv:2609.09830 give fewer than about 1.03 LZ events and need 9.4–65.5 t-y.

Two points of tension remain:

- **Splitting value.** The brief's 340 MeV charged–neutral splitting differs from the 356 MeV one-loop asymptote quoted here. The critic suggested that a two-loop value of about 341 MeV may reconcile them, but this was not read.
- **Robustness of the cancellation.** Whether the elastic cancellation is robust is unresolved. arXiv:2607.20588 suggests NLO effects can drive the rate lower still.

Dead ends:

- **Limit figures.** Figure extraction from arXiv:2410.17036 and arXiv:2303.14729 returned no limit plots, only data-distribution and calibration figures. The LZ text region read had no 1 TeV value.
- **Abstract search.** An abstract search for recent thermal-mass papers returned unrelated LEP notes. Direct arXiv-ID lookup proved more effective.

Limitations

- **No computation.** No compute tool existed, so no relic density, event rate or reach map was computed. The m^2 table is an interpolation from one point with an inferred mass.
- **Thermal-mass spread.** Only one thermal-mass determination was available. The bound-state-inclusive value of arXiv:2205.04486 was not read, so the spread across calculations is unknown.
- **Direct-detection numbers not read.** The LZ and XENONnT limits at 1 TeV, the neutrino-fog value at 1 TeV, and next-generation projections (XLZD, arXiv:2410.17137) were not read. The headline ratio of about 60 or more remains an extrapolation.
- **δ_m0 formula.** The tree-level δ_m0 formula is unsourced.
- **Indirect detection.** Indirect detection was not addressed.
- **Bibliography.** Several quotes (Planck, the Chen and Hill passages, the Table III/VI/VII traces) are recorded on the board but not cited into references.bib.
- **Budget.** Both waves were used.

Open directions

1. Read the Y = 1/2 doublet thermal mass and its uncertainty from arXiv:2205.04486, to give a second, bound-state-inclusive determination to compare with about 1.14 TeV.

2. Obtain the LZ 2024 and XENONnT SI limits at 1 TeV from HEPData or the remaining paper text, replacing the linear extrapolation.

3. Read the XLZD Design Book (arXiv:2410.17137) for the xenon neutrino-fog value and the projected SI sensitivity at about 1 TeV.

4. Confirm in the text of arXiv:1411.6930 which mass the pure-higgsino values 0.126/0.127 refer to.

5. Identify Ref. [28] of arXiv:2609.09830 and quote its Eq. (9), to map δ_m0 against (μ, M1, M2) with explicit sign dependence.

6. Add one sourced sentence on indirect-detection (for example H.E.S.S. Galactic Center) limits on TeV higgsinos.

7. Provide a relic-density compute tool (for example micrOMEGAs) so that Omega h^2(m) at 0.9–1.3 TeV can actually be computed.

System note (2026-10-16 11:17:20)

Report generated: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/reports/261016_111404/report.pdf Title: The Thermal Pure Higgsino near 1.1 TeV versus Planck 2018, LZ 2024 and XENONnT: A Literature-Based Assessment Supplement (implementation, tool-call log, board, extra figures, code index): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/reports/261016_111404/supplement.pdf Transcript: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/reports/261016_111404/transcri
Folder: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/reports/261016_111404
(report.tex, supplement.tex, transcript.tex, figures/, bundle.json)

User (2026-10-22 11:02:44)

Survey recent arXiv work on the higgsino thermal relic target (1.0-1.2 TeV). Use the inelastic-DM tools to compute the relic density at several masses; compare with LZ/XENONnT sensitivities and the Planck abundance. Where does the higgsino sit relative to current limits?

Assistant (research mode, 2026-10-22 11:02:44)

Plan:

1. (closed) Use the serve-idm.sh tools to compute the pure-higgsino thermal relic density Omega h^2 at m_chi = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, find the mass where Omega h^2 = 0.120, and run the combined constraint check at the thermal point.

2. (closed) Compute the inelastic Z-mediated recoil spectrum, the LZ event expectation and the (m_chi, delta_m) constraint map for the thermal higgsino. Reproduce or test the 2609.01590 preferred region (delta_m = 481-519 keV, m_chi = 1000-1183 GeV) against the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines.

3. (blocked) Close the remaining literature gaps: the 2205.04486 doublet thermal mass, the LZ 2024 and XENONnT SI limits near 1 TeV, and verbatim citations for Planck 2018 and Chen and Hill.

4. (closed) Extract verbatim, citable inputs from the 2026 LZ-event papers: the halo models, rates and preferred regions, the neutrino exclusion lines, the radiative corrections and the rescue scenarios.

5. (closed) Run an analytic and tool-based kinematic check of delta_max(m_chi, v_max) at E_R = 248 keV. Identify the scan points compatible with the LZ event, and reconcile the tool's 776 km/s cutoff with the upper delta edges of 2609.01590 (profiled Maxwellian 422/457 keV; Halo 13 519/529 keV).

6. (blocked) Recover quoted evidence for the contested literature numbers (f-024, f-025, f-026, f-027), and replace the tool's unphysical p-wave indirect-detection pass with an s-wave higgsino comparison at 1.1 TeV.

Tool calls: `list_agent_files`, `list_input_files`, `list_datasets`, `list_files`, `get_dispatch`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `get_arxiv_metadata`, `get_arxiv_metadata`, `get_arxiv_metadata`, `search_arxiv_abstracts`, `search_arxiv`, `search_arxiv_abstr` `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `describe_tools`, `download_arxiv_full_text`, `compute_relic_density`, `compute_relic_density`, `compute_relic_density`, `compute_relic_density`, `compute_relic_density`, `download_arxiv_full_text`, `describe_tools`, `predict_recoil`, `predict_recoil`, `predict_recoil`, `predict_recoil`, `predict_recoil`, `read_arxiv_chunk`, `read_arxiv_chunk`, `compute_relic_density`, `compute_relic_density`, `read_arxiv_chunk`, `read_arxiv_chunk`, `check_all_constraints`, `check_all_constraints`, `check_all_constraints`, `get_dispatch`, `scan_mass_splitting`, `read_arxiv_chunk`, `read_arxiv_chunk`, `plot_recoil_spectrum`, `predict_annual_modulation`, `check_all_constraints`, `cite`, `cite`, `get_arxiv_metadata`, `get_arxiv_metadata`, `get_arxiv_metadata`, `post_finding`, `post_finding`, `plot_constraint_summary`, `post_finding`, `post_finding`, `post_finding`, `post_question`, `post_finding`, `post_dead_end`, `cite`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_question`, `cite`, `post_dead_end`, `load_table_data`, `describe_tools`, `scan_mass_splitting`, `predict_recoil`, `predict_recoil`, `predict_recoil`, `predict_recoil`, `download_arxiv_full_text`, `predict_recoil`, `predict_recoil`, `download_arxiv_full_text`, `cite`, `download_arxiv_full_text`, `cite`, `download_arxiv_fu` `list_files`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `cite`, `cite`, `predict_recoil`, `predict_recoil`, `plot_constraint_su` `post_finding`, `post_finding`, `post_dead_end`, `post_question`, `post_finding`, `post_finding`, `post_dead_end`, `post_finding`, `post_finding`, `post_finding`, `post_dead_end`

Survey — 37 tools on 3 servers, 5 read-only probes, 13 papers found.

The pure higgsino, a Dirac-like electroweak doublet whose neutral states are split by a small Majorana mass, is a standard WIMP target. Its mass is fixed by requiring the thermal freeze-out abundance to match the Planck value Omega h^2 of about 0.12. That puts it near 1.0-1.2 TeV; the commonly quoted 1.1 TeV is background knowledge, and none of the returned abstracts state it. Sommerfeld enhancement and coannihilation with the charged state are needed to compute this mass precisely (1411.6930, 1008.2905). Bound-state effects are claimed to be important mainly above a few TeV (1407.7874). For elastic spin-independent scattering, the pure higgsino lies below the LZ 4.2 tonne-year limits (2410.17036; best SI limit 2.2e-48 cm^2, set at 40 GeV) and the XENONnT limits (2303.14729), because tree-level Z exchange is turned off by the inelastic mass splitting delta_m. The cross-section limits near 1.1 TeV are not given in these abstracts and must be read from the published curves. A burst of 2026 papers concerns a single LZ nuclear-recoil candidate at E_R = 248 keV (3.4 sigma local, 2.6 sigma global, from a 2.84 tonne-year exposure). It has been read as higgsino inelastic scattering with delta_m around 480-520 keV for m_chi = 1000-1183 GeV, which is exactly the thermal relic window (2609.01590). Neutrino constraints from dark matter captured and annihilating in the Sun are in tension with this reading: Super-K plus IceCube exclude delta_m below about 557 keV, which disfavors it for standard halo velocities (2609.07807), while IceCube alone excludes below about 475 keV once velocity effects near the blind spot are included (2609.01590). Other papers try to rescue the window (2610.03692) or look at boosted, non-halo explanations (2609.06890). Summary of where

the higgsino stands: the thermal 1.0-1.2 TeV higgsino is not excluded by elastic xenon limits. It is directly testable through inelastic Z-mediated scattering only for delta_m up to a few hundred keV, where LZ's high-recoil sensitivity, solar-neutrino limits and the CTAO gamma-ray line together close in on the remaining parameter space.

Quantities worth checking: Omega h^2 versus m_chi for a pure higgsino at about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, with and without Sommerfeld enhancement and coannihilation; find the mass where Omega h^2 = 0.12 (Planck); Sensitivity of the relic mass to the charged-neutral splitting (about 340 MeV radiative) and to the neutral splitting delta_m; Elastic SI cross section of the pure higgsino (loop- and Higgs-mediated) at 1-1.2 TeV, compared with the LZ 4.2 tonne-year and XENONnT 90% CL curves at the same mass; Inelastic Z-mediated scattering rate and recoil spectrum in xenon as a function of delta_m (about 400-600 keV) and m_chi; minimum recoil energy E_R,min and the kinematic threshold v_min(delta_m); Expected LZ event count above 100 keV for the 2.84 or 4.2 tonne-year exposure, compared with the single 248 keV event; Dependence of the rate on v_esc and on the LMC-induced high-velocity tail.

Plan — 4 lines of investigation:

1. Use the serve-idm.sh tools to compute the pure-higgsino thermal relic density Omega h^2 at m_chi = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, find the mass where Omega h^2 = 0.120, and run the combined constraint check at the thermal point. — The task asks for relic densities computed with the inelastic-DM tools. Earlier directions d-01 and d-04 could only interpolate from one published point in 1411.6930 because no relic-density tool was connected at the time. compute_relic_density and check_all_constraints are now in the catalog, so this direction answers the core question directly. It also replaces the low-confidence m^2 rescaling (f-009, about 1.14 TeV) with tool output.

2. Compute the inelastic Z-mediated recoil spectrum, the LZ event expectation and the (m_chi, delta_m) constraint map for the thermal higgsino. Reproduce or test the 2609.01590 preferred region (delta_m = 481-519 keV, m_chi = 1000-1183 GeV) against the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines. — Inelastic scattering is the only direct-detection route for the higgsino at current exposures. There is an unresolved tension here. The hand calculation f-001 gives a kinematic delta_max of about 364-368 keV for v_max = 776 km/s. 2609.01590, however, fits delta_m of about 500 keV, which needs a high-velocity tail (from the LMC) or a different halo model. The tool calculation must settle what delta_m can still be reached, and under which halo assumptions.

3. Close the remaining literature gaps: the 2205.04486 doublet thermal mass, the LZ 2024 and XENONnT SI limits near 1 TeV, and verbatim citations for Planck 2018 and Chen and Hill. — The tool results from d-09 and d-10 need published benchmarks. The thermal mass needs a second independent calculation (q-08). The elastic comparison still rests on a linear extrapolation of the 40 GeV limit (f-008, f-011), and the Planck and Chen-Hill quotes are on the board but not yet in references.bib.

4. Extract verbatim, citable inputs from the 2026 LZ-event papers: the halo models, rates and preferred regions, the neutrino exclusion lines, the radiative corrections and the rescue scenarios. — No direction reads 2609.01590, 2609.07807, 2610.03692, 2609.09830 or 2609.06890. d-10 relies on their numbers (481-519 keV, 475 and 557 keV, the LMC tail), and the solar-capture cross-check in the quantities list has no tool. The reason the two neutrino bounds differ (halo velocity tail, capture treatment) can only come from the literature. (added by the critic) Critic's view: revise — The steps need about 17 calls: 1 describe, 5 masses, 3 toggles, up to 3 bracketing, 4 sensitivity, 1 constraint check. The budget is 15. Drop the 400 and 600 keV delta_m variations, since a keV-scale neutral splitting has a negligible effect on freeze-out. Keep one delta_m variation as a sanity check

and use the 2 freed calls for the charged-splitting variations, which do matter through coannihilation.; If describe_tools shows that compute_relic_density is perturbative only, or omits coannihilation, the crossing will land near 1.0 TeV or lower. It must then not be reported as the higgsino thermal mass. State the result as tool output under those stated assumptions, and compare it with 1411.6930 and 2205.04486 (from d-11) rather than replacing them.; The 1172 GeV value for 1411.6930 is inferred, not stated in the paper. Label it as inferred everywhere. For check_all_constraints, record whether its LZ test is elastic SI, inelastic, or both, and which exposure it assumes. Without this, the LZ verdict cannot be interpreted..

Wave 1 — now exploring, in parallel: Use the serve-idm.sh tools to compute the pure-higgsino thermal relic density Omega h^2 at m_chi = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, find the mass where Omega h^2 = 0.120, and run the combined constraint check at the thermal point.; Compute the inelastic Z-mediated recoil spectrum, the LZ event expectation and the (m_chi, delta_m) constraint map for the thermal higgsino. Reproduce or test the 2609.01590 preferred region (delta_m = 481-519 keV, m_chi = 1000-1183 GeV) against the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines.; Extract verbatim, citable inputs from the 2026 LZ-event papers: the halo models, rates and preferred regions, the neutrino exclusion lines, the radiative corrections and the rescue scenarios..

Finding: At m_chi=1100 GeV (higgsino, sigma_n=7.43e-39 cm^2, hard halo cutoff v_esc+v_E=776 km/s), predict_recoil gives v_min(248 keV)=701/823/921/969/1018 km/s and E_R,peak=270/360/432/468/504 keV for delta=300/400/480/520/560 keV, with N_LZ(extended NR, 2.84 t-yr)=744.6 at 300 keV and 0.00 for delta>=400 keV; the 2609.01590 window (481-519 keV) gives zero events under the tool halo, consistent with f-001 delta_max of about 365 keV.

Finding: Local scan_mass_splitting (dispatch reported 'local execution') over m_chi=900-1300 GeV, delta=300-650 keV (15x15) finds 180/225 points pass all constraints and only 12 predict 0.3-5 LZ events; check_all_constraints at (1100 GeV, 500 keV) gives Omega h^2=0.1083, N_LZ=0.00 (not consistent with 1 event), sigma_v=1.29e-41 cm^3/s vs Fermi 2.3e-25, so 'viable' only because it is invisible; annual modulation at 248 keV is 0 (no rate).

Finding: Tool compute_relic_density (perturbative Kolb-Turner freeze-out, higgsino default sigma_v = 2.5e-26 cm^3/s x (1100 GeV/m_chi)^2, x_f about 24.5, g* = 86.25) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285, 0.1503 at m_chi = 900, 1000, 1100, 1200, 1300 GeV, and 0.1181 (1150 GeV) and 0.1202 (1160 GeV), so the thermal mass Omega h^2 = 0.120 is at about 1159 GeV (bracketed 1150-1160 GeV).

Finding: The serve-idm.sh relic tool is perturbative only: it has no Sommerfeld or coannihilation switch, no charged-splitting input and no delta_m input (Omega h^2 = 0.1202 at 1160 GeV for delta = 400, 500 and 600 keV). Its 'consistent_planck' flag is unreliable: it reports True at 25 sigma tension (1000 and 1300 GeV) and False at 39 sigma (900 GeV).

Finding: check_all_constraints at m_chi = 1160 GeV returns 'VIABLE' for delta = 400, 500 and 600 keV, with relic Omega h^2 = 0.1202 (0.1 sigma). It gives N_expected_LZ = 0.00 (2.84 t-yr, sigma_n = 7.43e-39 cm^2), which is inconsistent with the 1 observed event. Indirect sigma_v_today = 6.7e-42, 1.04e-41 and 1.50e-41 cm^3/s, against a Fermi limit of 2.48e-25 cm^3/s. CMB p_ann = 0.86, 1.35 and 1.94 e-45, against a limit of 3.2e-28.

Finding: plot_constraint_summary on the 900-1300 GeV x 300-650 keV scan returns best fit m_chi=986 GeV, delta=350 keV (plot /Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png); this lies below both the 475 keV (IceCube, 2609.01590) and 557 keV (Super-K+IceCube, 2609.07807) solar-neutrino exclusion lines, so under the tool's truncated SHM (v_max=776 km/s) no point reproducing the LZ event survives the neutrino limits, and the 2609.01590 region (481-519 keV) requires a high-velocity tail (e.g. LMC) not modelled by the tool.

Completed — Use the serve-idm.sh tools to compute the pure-higgsino thermal relic density Omega h^2 at m_chi = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, find the mass where Omega h^2 = 0.120, and run the combined constraint check at the thermal point.. Done. compute_relic_density (serve-idm.sh) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285 and 0.1503 at m_chi = 900, 1000, 1100, 1200 and 1300 GeV. It gives 0.1181 at 1150 GeV and 0.1202 at 1160 GeV, so the thermal mass (Omega h^2 = 0.120, Planck 0.12 +/- 0.0012) is about 1159 GeV, bracketed between 1150 and 1160 GeV (f-020). This is about 1 percent below the 1172 GeV inferred from 1411.6930 and about 2 percent above the f-009 estimate of about 1.14 TeV. The agreement is by construction, not from independent physics. The tool does perturbative Kolb-Turner freeze-out (x_f about 24.5, g* = 86.25) with sigma_v = 2.5e-26 cm^3/s x (1100 GeV/m_chi)^2. It has no Sommerfeld or coannihilation switch, no charged-splitting input and no delta_m dependence: Omega h^2 = 0.1202 at delta = 400, 500 and 600 keV (f-021). So step 3 (toggling Sommerfeld/coannihilation) and the charged-splitting part of step 5 could not be run. The tool's consistent_planck flag is unreliable: it reads True at 25 sigma tension and False at 39 sigma. check_all_constraints at 1160 GeV, delta = 500 keV returns 'VIABLE'. Relic is 0.1202 (0.1 sigma). LZ gives N_expected = 0.00 for 2.84 t-yr with sigma_n = 7.43e-39 cm^2, flagged inconsistent with the 1 observed event; no LZ limit value is returned. Fermi-LAT: sigma_v_today = 1.04e-41 cm^3/s against a limit of 2.48e-25. CMB: p_ann = 1.35e-45 against a limit of 3.2e-28. Results at 400 and 600 keV are similar (f-022). Warning: the tool treats annihilation today as p-wave and splitting-suppressed, at odds with the s-wave higgsino WW/ZZ rate of about 1e-26 cm^3/s, so its indirect and CMB passes are not physically reliable (q-15). No files or figures were produced.

Completed — Compute the inelastic Z-mediated recoil spectrum, the LZ event expectation and the (m_chi, delta_m) constraint map for the thermal higgsino. Reproduce or test the 2609.01590 preferred region (delta_m = 481-519 keV, m_chi = 1000-1183 GeV) against the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines.. All runs were local; get_dispatch reported 'local execution'. The tool uses a standard halo with a hard cutoff at v_esc+v_E = 776 km/s, and none of its calls accept a v_esc or LMC-tail option. At m_chi = 1100 GeV (sigma_n = 7.43e-39 cm^2), predict_recoil gives these values for delta = 300/400/480/520/560 keV: peak E_R = 270/360/432/468/504 keV, v_min at the peak = 701/809/886/922/957 km/s, and v_min at 248 keV = 701/823/921/969/1018 km/s. Expected events in the LZ extended nuclear-recoil search (2.84 t-yr): 744.6 at 300 keV and 0.00 for every delta of 400 keV or more. Scaling by exposure gives about 1101 events at 4.2 t-yr for 300 keV, and still zero for the rest. The tool reports only the peak E_R, not a separate E_R,min, and gives no integrated rate above 100 keV. These results agree with f-001's kinematic limit of about 365 keV (f-018). At (1100 GeV, 500 keV), check_all_constraints gives Omega h^2 = 0.108 and N_LZ = 0, marked 'not consistent with 1 event'. Fermi and CMB pass by many orders of magnitude, and the annual modulation at 248 keV is zero. The 15x15 scan (m_chi 900-1300 GeV, delta 300-650 keV, file /Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz) finds 180 of 225 points pass all constraints, but only 12 predict 0.3-5 LZ events. Its best fit is m_chi = 986 GeV, delta = 350 keV, which lies below both the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines (f-019, f-023). The scan output does not list which points are in the 12-point group, so the claim that they all sit below the neutrino lines is inferred from the 1.1 TeV kinematics, not confirmed by the tool. Plots: constraint summary at /Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png and the 1100 GeV / 500 keV recoil spectrum at /Users/nramachandra/github-integration/inelastic-dm-server/output/recoil_spectrum.png; the tools return no figure numbers. Conclusion: under the tool's halo, delta of about 480-520 keV gives zero events, not O(1), so no part of the 2609.01590 region survives. Explaining the LZ event at about 500 keV needs a high-velocity tail (for example from the LMC), which the tool does not model. How the rate changes with v_esc could not be tested, because no tool exposes

halo parameters (dead end x-03).

Finding: 2609.01590 (Wu, Zhang, Zhu) finds LZ-preferred delta_m = 481-519 keV (1 sigma) and 475-529 keV (2 sigma) for m_chi = 1000-1183 GeV with the Halo 13 MW+LMC velocity distribution, but only 312-422 keV (1 sigma) and 280-457 keV (2 sigma), best fit m_chi = 1.09 TeV and delta_m = 366 keV, for a profiled Maxwellian. The text says the upper reach in delta_m is set by the kinematic endpoint v_esc + v_E. The relic mass range is 1000-1183 GeV (1 sigma) and 943-1230 GeV (2 sigma). LZ inputs: E_R = 248 +/- 23 stat +/- 23 sys keV and 2.84 tonne-year exposure.

Finding: The 475 keV and 557 keV neutrino lines differ because of how the captured population in the Sun is treated, not because of the halo. 2609.01590 simulates a non-thermal population on extended solar orbits at m_chi = 1.11 TeV with no elastic cooling (elastic blind spot) and 10-year IceCube data, using <sigma v>_eff of about 1.5e-26 cm^3/s with Sommerfeld; this gives delta_m >~ 475 keV excluded at 90% CL (466 keV with loose solar/nuclear inputs, about 454 keV with the Fe nuclear-response uncertainty). If elastic scattering allows thermalization and capture-annihilation equilibrium, the same paper gets delta_m >~ 553 keV. 2609.07807 assumes thermalization within the Sun after capture, a standard truncated Maxwellian (u0 = 220 km/s, v_esc = 544 km/s, v_sun = 232 km/s), rho = 0.4 GeV/cm^3, sigma_chi_n = 7.4e-39 cm^2, <sigma v> = 1.3e-26 cm^3/s, m_chi = 1.08 TeV and a 95% CL chi^2 on IceCube nu_mu (2016), IceCube nu_e and Super-K data. It excludes delta_m <~ 557 keV and states the interpretation could remain viable with an LMC tail or substructure.

Finding: Gamma-ray line status at about 1.1 TeV (2609.01590): at the joint best-fit mass the thermal higgsino line prediction is 0.815-0.900 of the H.E.S.S. observed limits (546 h), so it is close to but not excluded. The matched CTAO Alpha 500 h projection gives mu_gamma,95 = 1.69, which is above the thermal target. The estimate for the Omega configuration, 1.69/2 = 0.85, is below mu_gamma = 1, so the thermal target is within CTAO exclusion reach. Dwarf, CMB and ATLAS disappearing-track limits do not exclude the region.

Finding: Abstract-level only: 2610.03692 (Freese and Theodosopoulos) finds the 1.1 TeV thermal higgsino only marginally compatible with the LZ event and IceCube, and only with the LMC tail, no elastic solar cooling and no LZ high-energy sideband. Its compatible regions extend from about 2 TeV to hundreds of TeV, with delta about 350-525 keV. 2609.09830 computes one-loop EW corrections, with on-shell renormalization reducing delta_m and Higgs-exchange elastic and inelastic rates; the abstract gives no number for the size of the change. 2609.06890 proposes light boosted DM whose energy is set by the dark-sector spectrum, so it is independent of the halo tail.

Completed — Extract verbatim, citable inputs from the 2026 LZ-event papers: the halo models, rates and preferred regions, the neutrino exclusion lines, the radiative corrections and the rescue scenarios.. I read the full texts of 2609.01590 and 2609.07807 and only the abstracts of 2610.03692, 2609.09830 and 2609.06890. The validated citations are in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib (keys arxiv260901590 and arxiv260907807). Not every number has a validated quote: the cite tool rejected several 2609.01590 sentences (the v_esc + v_E endpoint sentence, the 0.815-0.900 H.E.S.S. ratio, the 553 keV equilibrium bound, the Maxwellian best fit and the relic mass ranges) because the PDF text garbles math symbols and en-dash ranges. Those numbers were read directly in the text (tc-...-32, -36, -44) but have no validated quote (dead end x-04).

1. Preferred region, 2609.01590 (f-024, Fig. 1). With the Halo 13 MW+LMC velocity distribution the LZ fit gives delta_m = 481-519 keV at 1 sigma and 475-529 keV at 2 sigma, best fit about 496 keV. With a profiled Maxwellian halo it gives 312-422 keV at 1 sigma and 280-457 keV at 2 sigma, best fit 1.09 TeV and 366 keV. The paper says the upper

limit on delta_m is set by the kinematic endpoint v_esc + v_E, not by the event-rate normalization. The relic density fixes m_chi = 1000-1183 GeV at 1 sigma and 943-1230 GeV at 2 sigma. LZ inputs are E_R = 248 +/- 23 stat +/- 23 sys keV and 2.84 tonne-year.

1. Why the neutrino lines are 475 and 557 keV (f-025; Fig. 2 of 2609.01590, Figs. 2-3 of 2609.07807). The difference comes from how the captured higgsinos are treated in the Sun, not from the halo model.

- 2609.01590, at m_chi = 1.11 TeV: if elastic scattering is negligible (the blind spot), the captured higgsinos never thermalize and stay on extended orbits, so they annihilate less. With <sigma v>_eff of about 1.5e-26 cm^3/s and ten years of IceCube data, this excludes delta_m up to about 475 keV at 90% CL. Loose solar and nuclear inputs lower this to 466 keV, and the Fe nuclear-response uncertainty to about 454 keV. If elastic scattering lets them reach equilibrium, the same paper finds delta_m >~ 553 keV.
- 2609.07807 assumes the captured higgsinos thermalize and uses a standard truncated Maxwellian halo: u0 = 220 km/s, v_esc = 544 km/s, v_sun = 232 km/s, rho = 0.4 GeV/cm^3. Other inputs are sigma_chi_n = 7.4e-39 cm^2, <sigma v> = 1.3e-26 cm^3/s and m_chi = 1.08 TeV, with a 95% CL fit to IceCube nu_mu, IceCube nu_e and Super-K data. It excludes delta_m below about 557 keV, which agrees with the 553 keV equilibrium result. It still allows rescue by an LMC tail or substructure.

1. Gamma-ray lines near 1.1 TeV, 2609.01590 (f-026, Fig. 3). The thermal prediction is 0.815-0.900 of the H.E.S.S. 546 h limits, which the paper calls close to, but not excluded by, the present limits. The CTAO Alpha 500 h projection, mu_gamma,95 = 1.69, falls short of the thermal target. The Omega configuration, estimated at 0.85, would reach it. Dwarf-galaxy, CMB and ATLAS disappearing-track limits do not exclude the region.

1. Rescue and alternative scenarios, abstracts only (f-027).

- 2610.03692: the 1.1 TeV thermal higgsino is only marginally compatible with the LZ event and IceCube, and only with the LMC tail, no elastic cooling in the Sun and no sideband constraint. Compatible regions run from about 2 TeV to hundreds of TeV, with delta about 350-525 keV.
- 2609.09830: one-loop electroweak corrections with on-shell renormalization reduce delta_m. The abstract gives no size for the change.
- 2609.06890: light boosted dark matter, which does not depend on the halo velocity tail.

Still open (q-16): how large the 2609.09830 radiative correction is, and the numbers for the four IceCube variants in 2610.03692.

Critic, wave 1 — continue: 5 findings stand up to the evidence, 5 questioned ('check_all_constraints at m_chi = 1160 GeV returns 'VIABLE' for delta = 400, 500 and 600 ke': The quoted excerpts of tc-...-38/39/40 stop at the 'cmb_energy_injection' key. Neither the CMB p_ann values (0.86, 1.35, 1.94 e-45 against 3.2e-28) nor the overall 'VIABLE' verdict appears in the evidence. The relic, LZ and Fermi numbers do match. Re-quote the full JSON before treating the CMB and verdict parts as supported.; '2609.01590 (Wu, Zhang, Zhu) finds LZ-preferred delta_m = 481-519 keV (1 sigma) and 475-529': Neither evidence excerpt contains the claimed numbers. tc-...-32 is the title and author page, and tc-...-36 is the Fig. 3 gamma-ray caption. Neither shows 481-519, 475-529, 312-422, 280-457, 366 keV, 1.09 TeV, 1000-1183 GeV or 943-1230 GeV.

The d-12 summary also says the cite tool rejected these sentences (x-04). The values may be correct, but they are not evidenced.; 'The 475 keV and 557 keV neutrino lines differ because of how the captured population in th': Only part of the claim is evidenced. tc-...-37 supports the LMC/substructure rescue statement, and tc-...-45 shows the chi^2 procedure. No excerpt contains 475, 466, 454, 553 or 557 keV, the halo parameters (220/544/232 km/s), <sigma v> = 1.3e-26 or 1.5e-26, or m_chi = 1.08/1.11 TeV. There is also a labelling issue: tc-...-45 sets the limit at Delta chi^2 = 2.71, which is the one-sided 95% (two-sided 90%) threshold. This should be stated explicitly when comparing with the 90% CL IceCube line of 2609.01590.; 'Gamma-ray line status at about 1.1 TeV (2609.01590): at the joint best-fit mass the therma': tc-...-36 shows only that Fig. 3 plots H.E.S.S. and CTAO Omega curves of mu_gamma over 950-1200 GeV. The 0.815-0.900 ratio, the 546 h exposure and mu_gamma,95 = 1.69 do not appear in any excerpt. It is also unclear whether '1.69/2 = 0.85' comes from the paper or is the agent's own estimate. tc-...-44 concerns solar thermalization and is not relevant to the gamma-ray claim.).

Issues: The central task question has not been answered quantitatively: where does the elastic SI cross section of the thermal higgsino sit relative to the LZ 4.2 t-yr and XENONnT 90% CL curves at 1.0-1.2 TeV? No finding gives either the higgsino SI value (loop and Higgs-mediated, around 1e-48 to 1e-49 cm^2 with possible cancellation) or the limit values at 1.1 TeV. d-11 is still open, and this comparison is the core of the headline answer.; The thermal mass of about 1159 GeV is set by the tool's hard-coded normalization (sigma_v = 2.5e-26 x (1100/m)^2). It is a calibration, not a computation. The report must not present it as an independent check of 1411.6930. The Sommerfeld/coannihilation step and the charged-splitting sensitivity in the brief were never done. A literature or analytic estimate (1008.2905 method) is needed if the task's 'with and without Sommerfeld' item is to be answered.; The tool's indirect-detection and CMB passes rest on a p-wave, splitting-suppressed annihilation today. This is unphysical for a higgsino, which annihilates s-wave to WW/ZZ at about 1-1.5e-26 cm^3/s. The 'VIABLE' label for indirect and CMB constraints should be replaced by a comparison of the s-wave rate with the Fermi dwarf and H.E.S.S. continuum and line limits at 1.1 TeV, together with f-026..

Next steps after wave 1 — This is the final wave, with 24 tool calls. Open direction d-11 carries over and runs first. It already covers the critic's top issue, the elastic SI position of the thermal higgsino against the LZ 4.2 t-yr and XENONnT limits near 1.1 TeV and the neutrino fog, which is the core of the task's answer. I am not adding a duplicate. The executor should treat d-11 as SI-first: the higgsino SI prediction with its band, the limit values at 1.0-1.2 TeV, and the ratio; the 2205.04486 mass and the Planck citation are secondary. I add two directions. The first is a tight kinematic reconciliation using the tool and analytic delta_max, which tests the robustness of the 'no surviving point' claim against v_max and the LMC tail. The second is a narrow-window quote recovery plus an s-wave indirect-detection cross-check, which resolves the contested f-024 to f-027 and replaces the unphysical p-wave pass. The task does not ask for facility runs, so none is scheduled. Sommerfeld toggling cannot be done with the tools; the report should cite 1411.6930 for it and describe the tool's thermal mass as a calibration, not an independent check.

New lines of investigation: Run an analytic and tool-based kinematic check of delta_max(m_chi, v_max) at E_R = 248 keV. Identify the scan points compatible with the LZ event, and reconcile the tool's 776 km/s cutoff with the upper delta edges of 2609.01590 (profiled Maxwellian 422/457 keV; Halo 13 519/529 keV).; Recover quoted evidence for the contested literature numbers (f-024, f-025, f-026, f-027), and replace the tool's unphysical p-wave indirect-detection pass with an s-wave higgsino comparison at 1.1 TeV..

Wave 2 — now exploring, in parallel: Close the remaining literature gaps: the 2205.04486 doublet thermal mass, the LZ 2024 and XENONnT SI limits near 1 TeV, and verbatim citations for Planck 2018 and Chen and Hill.; Run an analytic and tool-based kinematic check of delta_max(m_chi, v_max) at E_R = 248 keV. Identify the scan points compatible with the

LZ event, and reconcile the tool's 776 km/s cutoff with the upper delta edges of 2609.01590 (profiled Maxwellian 422/457 keV; Halo 13 519/529 keV).; Recover quoted evidence for the contested literature numbers (f-024, f-025, f-026, f-027), and replace the tool's unphysical p-wave indirect-detection pass with an s-wave higgsino comparison at 1.1 TeV..

Finding: 2205.04486 (Bottaro et al., Table I) gives the thermal mass of the Dirac-fermion Y=1/2 doublet (2_{1/2}, higgsino-like) as M_DM = 1.08 +/- 0.02 TeV, including Sommerfeld enhancement and bound-state formation, with Omega_DM h^2 = 0.11933 +/- 0.00091; the paper also states the 2_{1/2} DD cross section lies well below the neutrino floor. This independent value is about 7 percent below the serve-idm.sh perturbative 1159 GeV (f-020).

Finding: Chen and Hill (1912.07795) is cited in references.bib (key arxiv191207795) with the verbatim abstract sentence 'The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range.'; the earlier board phrasing 'less than 10^-48 cm^2 in the TeV mass regime' was not found verbatim in the extracted text.

Finding: 2609.09830 (Chatterjee et al.) gives no on-shell delta_m shift in the 300-600 keV range. Its benchmarks are tuned so the one-loop on-shell (CCN[n] scheme) neutral splitting is delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5 (|mu| = 0.61-1.40 TeV; all have Omega h^2 < 0.12, so the relic constraint is not imposed). The tree-to-one-loop shift can be read from the Table III masses: about 0.14 GeV tree splitting vs 0 at one loop for BP1 (739.55/-739.69 to 739.33/-739.33 GeV), i.e. O(10-100) MeV scale. This paper therefore does not bear on the 475-557 keV window.

Finding: 2610.03692 (Freese and Theodosopoulos) shows four IceCube treatments at 90% CL as lower limits on delta in Fig. 2: capture-annihilation equilibrium; SI inelastic cooling only; SI inelastic + SD elastic (sigma_SI,el = 0); SI inelastic + SD + SI elastic (sigma_SI,el = 4e-50 cm^2). It uses the W+W- annihilation-rate limits, an SHM with v0 = 220, v_sun = 232 and v_esc = 544 km/s, rho = 0.3 GeV/cm^3 and sigma_Hn = 7.43e-39 cm^2. For the solar bound it says the neutrino limits 'remain unchanged by adding the effects of the LMC'. For LZ, the SHM+LMC eta(v_min) is set to 0 above 950 km/s. The numerical delta values at 1.1 TeV under each treatment appear only in the figure, not in the text I read, so they are not quantified here.

Blocked — Close the remaining literature gaps: the 2205.04486 doublet thermal mass, the LZ 2024 and XENONnT SI limits near 1 TeV, and verbatim citations for Planck 2018 and Chen and Hill.. (The tool budget ran out before steps 3-6. The LZ 2024 (2410.17036) and XENONnT (2303.14729) SI limits near 1.1 TeV and the XLZD neutrino-fog level were not read, and the comparison table was not built (q-17). The Planck citation needs a shorter fragment found in the extracted text and a second cite attempt.) Two of the five gaps are closed; the other three are still open. (1) The doublet thermal mass from 2205.04486 (Bottaro et al., Table I) is quoted verbatim. For the Dirac-fermion Y=1/2 doublet it is M_DM = 1.08 +/- 0.02 TeV, computed with Sommerfeld enhancement and bound-state formation, normalised to Omega_DM h^2 = 0.11933 +/- 0.00091. The paper also says the doublet's direct-detection cross section 'lies well below the neutrino' floor. This is an independent result and sits about 7 percent below the tool's perturbative 1159 GeV (f-020). It is in references.bib under key arxiv220504486 (f-028). (2) Chen and Hill (1912.07795) is in references.bib under key arxiv191207795, with the abstract sentence 'The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range.' The board's phrase 'less than 10^-48 cm^2 in the TeV mass regime' does not appear verbatim in the extracted text, so it was not cited (f-029). (3) Planck 2018 (1807.06209) is not cited yet. The quote '0.120 \pm 0.001' was not found in the extracted text, and the part of the abstract that was read does not contain the Omega_c h^2 sentence (dead end x-05). Bib file: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib. Full text of 2205.04486: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-

dd/runs/261022_110244/2205.04486.txt. No figures were produced.

Blocked — Recover quoted evidence for the contested literature numbers (f-024, f-025, f-026, f-027), and replace the tool's unphysical p-wave indirect-detection pass with an s-wave higgsino comparison at 1.1 TeV.. (The tool budget ran out before the 2609.01590 and 2609.07807 text windows could be read and before the indirect-detection comparison could be done.) Two of the four goals are settled from full texts. The other two, plus the indirect-detection comparison, were not reached. (1) 2609.09830 does not give an on-shell delta_m shift for the 475-557 keV window (f-030). Its benchmark points are tuned to small one-loop on-shell splittings: delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5, at |mu| = 0.61-1.40 TeV, all with Omega h^2 < 0.12. Comparing its tree and one-loop masses suggests the shift is of order 0.1 GeV, but that is my reading of their mass table, not a number the paper states. So the paper is not relevant to rescuing the 475-557 keV window. (2) 2610.03692 defines four IceCube treatments, shown at 90% CL in its Fig. 2 (f-031): capture-annihilation equilibrium; SI inelastic cooling only; SI inelastic + SD elastic cooling; and SI inelastic + SD + SI elastic cooling with sigma_SI,el = 4e-50 cm^2. Its inputs: W+W- limits, standard halo with v0 = 220, v_sun = 232 and v_esc = 544 km/s, rho = 0.3 GeV/cm^3, and an LMC tail cut off at v_min = 950 km/s. The solar limits are stated not to change when the LMC is added. The delta values at 1.1 TeV for each treatment appear only in the figure, so they are not quantified. The paper's text calls the 1.1 TeV thermal higgsino only marginally compatible. Blocked items (dead end x-06): the tool budget ran out before narrow re-reads of 2609.01590 and 2609.07807. Findings f-024, f-025 and f-026 therefore remain contested, including whether the 1.69/2 CTAO Omega estimate is from the paper or from the agent. No s-wave <sigma v> versus H.E.S.S. ratio was computed, so the tool's p-wave pass (f-022) is still unreplaced. The full texts are in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/ as 2609.01590.txt, 2610.03692.txt and 2609.09830.txt. No cite calls were made.

Finding: Under the tool's truncated SHM (v_max = 776 km/s), the LZ-compatible band (0.3-5 events, 2.84 t-yr) for m_chi = 1000-1200 GeV is confined to delta of about 350 keV. predict_recoil gives N = 10.67/17.30 at delta = 340 keV (m_chi = 1000/1183 GeV), 1.13/2.90 at 350 keV (1000/1200 GeV), 0.03/0.158 at 360 keV (1000/1183 GeV) and 0.00/0.00 at 380 keV. A 16x16 scan (m_chi 1000-1200 GeV, delta 330-480 keV in 10 keV steps) reports exactly 16 points in the 0.3-5 band, best fit (1000 GeV, 350 keV). No LZ-compatible point has delta >= 475 keV, or even >= 360 keV.

Finding: Analytic kinematics (m_N = 122.0 GeV for Xe-131, E_R = 248 keV, sqrt(2 m_N E_R) = 0.8206 keV per km/s): delta_max = v_max*sqrt(2 m_N E_R) - m_N E_R/mu gives, at m_chi = 1000/1100/1183 GeV, 358.5/361.3/363.2 keV for v_max = 776 km/s, 378/381/383 keV for 800 km/s, 419/422/424 keV for 850 km/s and 460/463/465 keV for 900 km/s. This reproduces the tool's v_min(248 keV) to within 1 km/s (766 vs 765.7 km/s at 1000 GeV and 350 keV; 921/969/1018 km/s at 1100 GeV and 480/520/560 keV). The tool's 'about 365 keV' edge is the E_R-optimized endpoint delta_max = mu v_max^2/2 = 364/368/370 keV at 776 km/s, reached at E_R = mu*delta/m_N of about 330 keV, not at 248 keV.

Finding: Inverting the kinematics at m_chi = 1100 GeV gives the v_max (Earth frame) needed to reach a given delta with an event exactly at E_R = 248 keV: 893 km/s for 457 keV, 915 km/s for 475 keV, 968 km/s for 519 keV, 980 km/s for 529 keV and 1015 km/s for 557 keV (about +3 km/s at 1000 GeV, about -2 km/s at 1183 GeV). If E_R is free (at the optimal E_R = mu*delta/m_N of about 380-500 keV), the minimum is v_max = sqrt(2 delta/mu): 830 km/s for 422, 864 for 457, 881 for 475, 921 for 519, 930 for 529 and 954 km/s for 557 keV. All of these exceed the tool's SHM cutoff of 776 km/s by 55-240 km/s. So the 2609.01590 profiled-Maxwellian upper edges (422/457 keV) imply an effective endpoint of about 830-890 km/s. The Halo 13 edges (519/529 keV) need about 920-980 km/s of LMC-boosted tail. Covering delta >= 557 keV

(above the Super-K + IceCube line) needs at least about 950-1015 km/s, and delta >= 475 keV needs at least about 880-915 km/s. The higgsino reading therefore rests entirely on the halo's population above about 880 km/s in the Earth frame.

Completed — Run an analytic and tool-based kinematic check of delta_max(m_chi, v_max) at E_R = 248 keV. Identify the scan points compatible with the LZ event, and reconcile the tool's 776 km/s cutoff with the upper delta edges of 2609.01590 (profiled Maxwellian 422/457 keV; Halo 13 519/529 keV).. Main result: under the tool's halo, which cuts off at 776 km/s, no point that reproduces the LZ event has delta of 475 keV or more; the compatible band sits at about 350 keV. Everything ran locally. Points in the 0.3-5 event band were identified by bracketing with predict_recoil, because the scan file could not be read (x-07).

Event counts from predict_recoil (LZ, 2.84 t-yr):

- delta 340 keV: 10.67 events at 1000 GeV and 17.30 at 1183 GeV
- delta 350 keV: 1.13 at 1000 GeV and 2.90 at 1200 GeV
- delta 360 keV: 0.03 at 1000 GeV and 0.158 at 1183 GeV
- delta 380 keV: 0.00 at both masses

A new 16x16 scan (m_chi 1000-1200 GeV, delta 330-480 keV in 10 keV steps) passes all 256 points and puts 16 in the 0.3-5 band, best fit (1000 GeV, 350 keV). That the 16 are the delta = 350 keV row is inferred from the bracketing, not listed by the tool (f-032). This scan overwrote /Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz and constraint_summary.png. The tools return no figure numbers.

Analytic check (f-033, hand-computed): delta_max = v_max*sqrt(2 m_N E_R) - m_N E_R/mu at E_R = 248 keV, with m_N = 122 GeV. At 1000/1100/1183 GeV it gives 358.5/361.3/363.2 keV for 776 km/s, 378/381/383 keV for 800, 419/422/424 keV for 850 and 460/463/465 keV for 900 km/s. It matches the tool's v_min(248 keV) to within about 1 km/s. The tool's 'about 365 keV' edge is the endpoint when E_R is left free, mu v^2/2 = 364-370 keV at 776 km/s, reached near E_R of about 330 keV rather than at 248 keV.

Required v_max (f-034, hand-computed), at 1100 GeV with the event at 248 keV: 893 km/s for 457 keV, 915 for 475, 968 for 519, 980 for 529 and 1015 for 557 keV. With E_R left free the minimum is lower: 830 km/s for 422 keV, 864 for 457, 881 for 475, 921 for 519, 930 for 529 and 954 for 557 keV.

What this means for 2609.01590 and the LMC tail:

- The paper's profiled-Maxwellian upper edges (422/457 keV) imply an effective endpoint of about 830-890 km/s, compared with the tool's 776 km/s. This explains why the tool gives zero events at 400 keV.
- The Halo 13 edges (519/529 keV) need an LMC-boosted tail reaching about 920-980 km/s.
- Clearing the 475 keV IceCube line needs at least about 880-915 km/s.
- Clearing the 557 keV Super-K + IceCube line needs at least about 950-1015 km/s. The higgsino reading of the event therefore depends entirely on how many particles the halo has above about 880 km/s in the Earth frame.

Still open: I could not compare these numbers with the actual v_esc + v_E values of the 2609.01590 halos, because the paper's endpoint sentence has no validated quote (x-04).

Dead end x-07: load_table_data cannot read the binary npz file.

Critic, wave 2 — continue: 4 findings stand up to the evidence, 3 questioned ('2205.04486 (Bottaro et al., Table I) gives the thermal mass of the Dirac-fermion Y=1/2 dou': The quoted excerpts do not contain the claimed values. tc-...-86 is the title page. tc-...-91 shows Table I rows for 4_{1/2} through 12_{1/2} Dirac fermions and the complex-scalar 2_{1/2} (0.58 +/- 0.01), but not the Dirac-fermion 2_{1/2} row with 1.08 +/- 0.02 TeV. Neither 0.11933 +/- 0.00091 nor the 'well below the neutrino floor' sentence appears. The cite call tc-...-97 returns only 'Cited:' and does not show which fragment was validated. The value is probably correct, but the evidence posted does not show it.; '2609.09830 (Chatterjee et al.) gives no on-shell delta_m shift in the 300-600 keV range. I': Neither excerpt (tc-...-93, tc-...-94) contains the BP1-BP5 splittings (36.37/8.90/16.39/10.81/10.59 keV), the |mu| range, the Omega h^2 < 0.12 statement or the Table III masses. They contain only the counterterm text and the Fig. 4 caption. The tree-to-loop shift is also the agent's own reading. A table quoted to 0.01 GeV cannot resolve keV-scale one-loop splittings. A shift of about 140 MeV against a keV-scale delta also implies strong scheme sensitivity, which may bear on the 475-557 keV window rather than being irrelevant to it.; '2610.03692 (Freese and Theodosopoulos) shows four IceCube treatments at 90% CL as lower li': Only partly supported. The excerpts support the equilibrium case plus three other cooling scenarios, the 90% CL treatment and the 'marginal' reading at 1.1 TeV. None of the specific inputs appears in any excerpt: v0 = 220, v_sun = 232, v_esc = 544 km/s, rho = 0.3 GeV/cm^3, sigma_Hn = 7.43e-39 cm^2, the 950 km/s LMC eta cutoff, the 'remain unchanged by adding the effects of the LMC' quote and sigma_SI,el = 4e-50 cm^2.).

Issues: The core comparison asked for in the task is still missing after two waves. No LZ 4.2 t-yr or XENONnT SI 90% CL limit near 1.0-1.2 TeV has been read. The higgsino elastic sigma_SI has not been placed against those limits or the neutrino fog, and Planck 2018 is uncited. The final answer cannot be stated quantitatively without these.; The 'compatible band at about 350 keV' is an artefact of the tool's hard 776 km/s cutoff. N falls from 10.7 to 1.1 to 0.03 over 20 keV. This should be stated as a property of the truncated SHM, not as a physical preferred region. The scan also overwrote the d-10 scan_idm.npz and constraint_summary.png, so the f-019/f-023 outputs can no longer be reproduced from disk.; The kinematics assume a single isotope with m_N = 122 GeV (Xe-131). Natural xenon spans A = 124-136. For example, Xe-134 raises delta_max at 776 km/s and 1.1 TeV by about 7 keV. The 248 +/- 23 stat +/- 23 sys keV energy uncertainty also matters. The v_max thresholds therefore carry roughly +/- 10-20 km/s, which should be stated..

Stopping after wave 2: budget reached (2/2 waves used). Writing the report from the board.

Research report:

The thermal higgsino at 1.0–1.2 TeV: relic mass, inelastic LZ rates and neutrino limits

Abstract

This run surveyed recent work on the pure-higgsino thermal relic, used the inelastic-DM tool server to compute relic densities and xenon recoil rates, and checked the 2026 reading of the LZ 248 keV nuclear-recoil event as higgsino inelastic scattering.

- **Relic mass from the tool.** The tool's perturbative freeze-out gives Omega h^2 = 0.120 at m_chi ≈ 1159 GeV (bracketed by 1150 and 1160 GeV). This is a calibration of the tool's built-in cross-section normalisation, not an independent calculation. The independent Sommerfeld plus bound-state value from the literature is 1.08 ± 0.02 TeV.
- **LZ event under the tool's halo.** With a standard halo cut off at 776 km/s, the tool

predicts zero LZ events for every delta_m $\geq$ 380 keV at 1.0–1.2 TeV. LZ-compatible event counts appear only near delta_m $\approx$ 350 keV, which lies below both the 475 keV (IceCube) and 557 keV (Super-K + IceCube) solar-neutrino exclusions.

- **What the literature region requires.** Analytic kinematics show that the published preferred region (481–519 keV) needs Earth-frame dark-matter speeds of about 920–980 km/s, and clearing the 557 keV line needs about 950–1015 km/s. The higgsino reading of the event therefore depends entirely on the halo's high-velocity tail.
- **Not done.** The elastic spin-independent comparison against the LZ and XENONnT curves at 1.1 TeV was not completed.

Background

The pure higgsino is an electroweak doublet whose neutral states are split by a small Majorana mass, delta_m. Its thermal mass is fixed by matching the Planck abundance, Omega h^2 $\approx$ 0.12. A precise value needs Sommerfeld enhancement and coannihilation with the charged state (arXiv:1411.6930; analytic method in arXiv:1008.2905). Bound-state formation is claimed to matter mainly above a few TeV (arXiv:1407.7874).

Bottaro et al. (arXiv:2205.04486, Table I) give M_DM = 1.08 $\pm$ 0.02 TeV for the Dirac Y=1/2 doublet. That value includes Sommerfeld enhancement and bound-state formation and is normalised to Omega_DM h^2 = 0.11933 $\pm$ 0.00091. The same paper states that the doublet's direct-detection cross section "lies well below the neutrino" floor. Chen and Hill (arXiv:1912.07795) state: "The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range." The relic density also sets the upper mass limit relevant to collider disappearing-track searches (arXiv:1901.02987).

For elastic scattering, the higgsino lies below the LZ 4.2 tonne-year limits (arXiv:2410.17036; best SI limit 2.2e-48 cm^2 at 40 GeV) and the XENONnT limits (arXiv:2303.14729). Tree-level Z exchange is removed by the inelastic splitting.

The LZ event. LZ reported a nuclear-recoil candidate at E_R = 248 $\pm$ 23 stat $\pm$ 23 sys keV, with 3.4σ local and 2.6σ global significance, in a 2.84 tonne-year exposure (arXiv:2609.06890, arXiv:2609.02608).

Papers interpreting or constraining the event:

- Wu, Zhang and Zhu (arXiv:2609.01590) fit the event with a higgsino.
- With a Milky Way plus Large Magellanic Cloud (LMC) velocity distribution ("Halo 13"), they find delta_m = 481–519 keV at 1σ and 475–529 keV at 2σ.
- With a profiled Maxwellian halo they find 312–422 keV at 1σ and 280–457 keV at 2σ, with best fit 1.09 TeV and 366 keV.
- The relic density fixes m_chi = 1000–1183 GeV at 1σ and 943–1230 GeV at 2σ.
- Solar-capture neutrino limits:
- arXiv:2609.07807 excludes delta_m ? 557 keV, assuming captured higgsinos thermalise in a standard truncated Maxwellian halo.
- arXiv:2609.01590 excludes delta_m ? 475 keV at 90% CL with IceCube when captured particles stay on extended orbits without elastic cooling. Loose inputs give 466 keV and the Fe nuclear-response uncertainty gives about 454 keV. Assuming equilibrium instead, the same paper finds about 553 keV.

- Freese and Theodosopoulos (arXiv:2610.03692) find the 1.1 TeV thermal higgsino only marginally compatible with the event and IceCube. Their compatible regions run from about 2 TeV to hundreds of TeV with delta ≈ 350–525 keV.
- arXiv:2609.09830 computes one-loop electroweak corrections to delta_m and to the scattering rates.
- arXiv:2609.06890 proposes boosted dark matter, which does not depend on the halo tail.

Several of these literature numbers were read directly in the full texts but could not be validated as verbatim quotes, because PDF extraction garbles math symbols and en-dash ranges. Treat them as read but not independently verified.

Results

Relic density versus mass (tool) The tool's `compute_relic_density` uses perturbative Kolb–Turner freeze-out (x_f ≈ 24.4–24.8, g* = 86.25) with sigma_v = 2.5e-26 cm^3/s × (1100 GeV/m_chi)^2.

m_chi (GeV)	Omega h^2
900	0.0731
1000	0.0898
1100	0.1083
1150	0.1181
1160	0.1202
1200	0.1285
1300	0.1503

The Planck crossing (0.120 ± 0.0012) is at about 1159 GeV.

- **Comparison with the literature.** This is about 1% below the 1172 GeV inferred, not quoted, from arXiv:1411.6930. It is about 7% above the 1.08 TeV of arXiv:2205.04486.
- **Why it is not a check.** The agreement follows from the hard-coded normalisation; it is not independent physics.
- **Missing inputs.** The tool has no Sommerfeld or coannihilation switch, no charged-splitting input and no delta_m dependence: Omega h^2 = 0.1202 at 1160 GeV for delta = 400, 500 and 600 keV. The "with and without Sommerfeld/coannihilation" and charged-splitting sensitivity items could not be run.
- **Unreliable flag.** The tool's `consistent_planck` flag reads True at 25σ tension and False at 39σ.

Inelastic recoil kinematics and LZ event counts (tool) The tool's halo is a standard halo with a hard cutoff at v_esc + v_E = 776 km/s. No call exposes halo parameters. Results at m_chi = 1100 GeV, sigma_n = 7.43e-39 cm^2:

delta (keV)	E_R,peak (keV)	v_min at peak (km/s)	v_min at 248 keV (km/s)	N_LZ (2.84 t-yr)
300	270	701	701	744.6

delta (keV)	E_R,peak (keV)	v_min at peak (km/s)	v_min at 248 keV (km/s)	N_LZ (2.84 t-yr)
400	360	809	823	0.00
480	432	886	921	0.00
520	468	922	969	0.00
560	504	957	1018	0.00

Scaling by exposure gives about 1101 events at 4.2 t-yr for 300 keV and zero for the rest. The tool does not report a separate E_R,min or an integrated rate above 100 keV. The annual-modulation output at (1100 GeV, 500 keV, 248 keV) is identically zero, because the rate is zero.

Bracketing near the edge with `predict_recoil` (2.84 t-yr):

delta (keV)	events at 1000 GeV	events at upper mass
340	10.67	17.30 (1183 GeV)
350	1.13	2.90 (1200 GeV)
360	0.03	0.158 (1183 GeV)
380	0.00	0.00 (1183 GeV)

Scans.

- A 15×15 scan (900–1300 GeV, 300–650 keV) found 180 of 225 points passing all constraints. Only 12 points predict 0.3–5 LZ events, with best fit (986 GeV, 350 keV).
- A 16×16 scan (1000–1200 GeV, 330–480 keV) passed all 256 points. It placed 16 points in the 0.3–5 band, with best fit (1000 GeV, 350 keV).
- That those 16 points form the delta = 350 keV row is inferred from the bracketing above. The binary scan file could not be read, so this was not listed directly by the tool.
- The second scan overwrote the first scan's file (`.../inelastic-dm-server/output/scan_idm.npz`) and its summary plot (`constraint_summary.png`). A recoil-spectrum plot for 1100 GeV / 500 keV is at `.../output/recoil_spectrum.png`.
- No numbered figures were produced.

Interpretation. Under the truncated halo, no LZ-compatible point reaches delta $\geq$ 360 keV. In particular none reaches the 475 or 557 keV neutrino lines, so no tool point that reproduces the event survives the neutrino limits. The critic stressed that the band near 350 keV is an artefact of the hard cutoff, where N falls from 10.7 to 1.1 to 0.03 over 20 keV. It is not a physical preferred region.

Analytic kinematic check The maximum splitting that can produce a recoil at E_R = 248 keV for a given maximum speed is

delta_max = v_max·sqrt(2 m_N E_R) - m_N E_R/mu,

with m_N = 122.0 GeV (Xe-131) and sqrt(2 m_N E_R) = 0.8206 keV per km/s.

- **Agreement with the tool.** This reproduces the tool's v_min(248 keV) to within about 1 km/s.

- **Origin of the "about 365 keV" edge.** That figure is the endpoint with E_R left free, mu v_max^2/2 = 364–370 keV at 776 km/s, reached near E_R ≈ 330 keV rather than at 248 keV.

v_max (km/s)	**delta_max at 1000 GeV (keV)**	**at 1100 GeV**	**at 1183 GeV**
776	358.5	361.3	363.2
800	378	381	383
850	419	422	424
900	460	463	465

Inverting at 1100 GeV gives the Earth-frame v_max needed to reach a given splitting:

delta (keV)	**v_max with event at 248 keV (km/s)**	**v_max with E_R free (km/s)**
422	—	830
457	893	864
475	915	881
519	968	921
529	980	930
557	1015	954

At 1000 GeV these values shift by about +3 km/s, and at 1183 GeV by about -2 km/s.

What this implies:

- The profiled-Maxwellian edges of arXiv:2609.01590 (422/457 keV) imply an effective endpoint of about 830–890 km/s.
- The Halo 13 edges (519/529 keV) need an LMC-boosted tail reaching about 920–980 km/s.
- Clearing 475 keV needs at least about 880–915 km/s, and clearing 557 keV needs at least about 950–1015 km/s.

The critic noted two sources of uncertainty that make these thresholds uncertain by roughly ±10–20 km/s:

- The calculation assumes a single isotope. Natural xenon spans A = 124–136, and Xe-134 raises delta_max by about 7 keV at 776 km/s and 1.1 TeV.
- The ±23 keV statistical and systematic energy uncertainty also enters.

Combined constraint check (tool) At 1160 GeV, `check_all_constraints` returns "VIABLE" for delta = 400, 500 and 600 keV:

Quantity	**delta = 400 keV**	**500 keV**	**600 keV**	**Tool's limit**
Omega h^2	0.1202 (0.1σ)	0.1202	0.1202	—
N_LZ (2.84 t-yr)	0.00	0.00	0.00	flagged inconsistent with 1 observed event

Quantity	delta = 400 keV	500 keV	600 keV	Tool's limit
sigma_v today (cm^3/s)	6.7e-42	1.04e-41	1.50e-41	2.48e-25 (Fermi)
CMB p_ann	0.86e-45	1.35e-45	1.94e-45	3.2e-28

At (1100 GeV, 500 keV) the check gives Omega h^2 = 0.1083, N_LZ = 0 and sigma_v = 1.29e-41 against a Fermi limit of 2.3e-25.

Warning: these verdicts are not physically reliable. The tool treats annihilation today as p-wave and splitting-suppressed, whereas the higgsino annihilates s-wave to WW/ZZ at about 1e-26 cm^3/s. The "viable" label therefore means invisible, not allowed. The critic contested this finding, and no s-wave replacement was computed.

Literature: why the two neutrino lines differ, and gamma-ray lines The 475 vs 557 keV difference comes from how captured particles are treated in the Sun, not from the halo model (read in the full texts; not independently verified).

- **arXiv:2609.01590 (475 keV).** At 1.11 TeV it uses a non-thermal captured population on extended orbits, <sigma v>_eff ≈ 1.5e-26 cm^3/s with Sommerfeld, and ten years of IceCube data.
- **arXiv:2609.07807 (557 keV).** It assumes thermalisation, a truncated Maxwellian (u0 = 220, v_esc = 544, v_sun = 232 km/s; rho = 0.4 GeV/cm^3), sigma_chi_n = 7.4e-39 cm^2, <sigma v> = 1.3e-26 cm^3/s and m_chi = 1.08 TeV. It uses a 95% CL fit to IceCube nu_mu, IceCube nu_e and Super-K. It still allows rescue by an LMC tail or substructure.
- **arXiv:2610.03692.** It shows four 90% CL IceCube cooling treatments in its Fig. 2:
- capture–annihilation equilibrium;
- spin-independent (SI) inelastic cooling only;
- SI inelastic plus spin-dependent (SD) elastic cooling;
- SI inelastic plus SD plus SI elastic cooling, with sigma_SI,el = 4e-50 cm^2.

Its other inputs are a standard halo with rho = 0.3 GeV/cm^3, sigma_Hn = 7.43e-39 cm^2 and an LMC tail cut at 950 km/s. It states the solar limits "remain unchanged by adding the effects of the LMC". The delta values at 1.1 TeV for each treatment appear only in the figure and were not quantified.

Gamma-ray lines (arXiv:2609.01590). The thermal line prediction is 0.815–0.900 of the H.E.S.S. 546 h limits, which the paper describes as close to but not excluded. The CTAO Alpha 500 h projection gives mu_gamma,95 = 1.69, which falls short of the thermal target. An Omega-configuration estimate of 0.85 would reach it, but it is unverified whether that number comes from the paper or from the agent's own 1.69/2 scaling. Dwarf, CMB and ATLAS disappearing-track limits do not exclude the region.

Radiative corrections (arXiv:2609.09830). Its benchmarks have one-loop on-shell splittings of 36.37, 8.90, 16.39, 10.81 and 10.59 keV at |mu| = 0.61–1.40 TeV, all with Omega h^2 < 0.12. It gives no delta_m shift in the 475–557 keV window. A tree-to-one-loop shift of order 0.1 GeV was read off its mass table by the agent; the paper does not state it. It therefore does not bear on rescuing the window.

Discussion

Relic mass. The tool's 1159 GeV is a normalisation echo. The best independent anchor in the record is 1.08 ± 0.02 TeV (arXiv:2205.04486), which includes Sommerfeld enhancement and bound states. The 1172 GeV figure attributed to arXiv:1411.6930 is inferred rather than quoted. Which value to use as the headline remains open, since published thermal masses differ by roughly 10–20% depending on treatment.

The LZ event. Our computations and the literature agree on one point: the LZ reading requires a population of halo particles above roughly 880 km/s in the Earth frame. A standard halo truncated at 776 km/s caps the splitting at about 360 keV at 248 keV recoil. That is below every neutrino exclusion line. This is consistent with arXiv:2609.07807's statement that the reading is disfavoured for standard halo velocities, and with arXiv:2610.03692's "marginal" verdict.

What the run could not do is decide between the published analyses. They differ in the solar treatment (thermalised or not) and in how much LMC tail they assume. Our tool can vary neither.

Current limits. The task asked where the higgsino sits relative to current limits. Only qualitative literature statements are available for the elastic channel: below the LZ and XENONnT reach, and below the neutrino floor according to arXiv:1912.07795 and arXiv:2205.04486. No numerical limit at 1.1 TeV was read. For indirect detection, the only computed numbers come from the tool's unphysical p-wave treatment. The literature gamma-ray line ratio (0.815–0.900 of H.E.S.S.) suggests the thermal higgsino is close to current sensitivity, but that number is not independently verified.

Dead ends.

- No tool exposes halo parameters, so the dependence on v_esc and on an LMC tail could not be tested. A custom rate code is needed.
- `load_table_data` cannot read the binary scan file, so scan-band membership had to be inferred by bracketing.
- Verbatim quotes from arXiv:2609.01590 failed because of garbled math: the endpoint sentence, the H.E.S.S. ratio, the 553 keV bound, the Maxwellian best fit and the relic ranges.
- The Planck 2018 quote "0.120 \pm 0.001" was not found in the extracted text, so Planck is not yet cited in the reference file.
- The tool budget ran out before narrow re-reads of arXiv:2609.01590 and arXiv:2609.07807 and before an s-wave indirect-detection comparison.

Limitations

- **Relic density.** The tool is perturbative only, with no Sommerfeld, coannihilation, charged-splitting or delta_m dependence. The relic mass is therefore not an independent computation.
- **Event counts.** Counts assume the tool's unstated LZ acceptance in its extended nuclear-recoil search and a single halo with a hard 776 km/s cutoff. There is no isotope averaging, and the critic noted thresholds carry about ±10–20 km/s.

- **Elastic SI comparison.** Not done: the LZ 4.2 t-yr and XENONnT 90% CL values near 1.1 TeV, the higgsino sigma_SI band and the neutrino-fog level were never read. The critic flagged this as the main missing piece.
- **Indirect and CMB.** These rest on an unphysical p-wave assumption. No s-wave comparison was made.
- **Unverified literature numbers.** Several literature numbers (preferred regions, 553 keV, the H.E.S.S. ratio, the CTAO Omega estimate, the doublet mass, the arXiv:2610.03692 and arXiv:2609.09830 details) were not independently verified. The critic contested them, mostly because of failed verbatim quotes or abstract-only reading.
- **Reproducibility.** The first scan's outputs were overwritten and cannot be reproduced from disk.
- **Compute.** All computation ran locally. No HPC facility was used.

Open directions

1. Read the LZ 4.2 t-yr (arXiv:2410.17036) and XENONnT (arXiv:2303.14729) SI 90% CL limits near 1.1 TeV from text or HEPData and place the Chen–Hill higgsino band and the neutrino fog against them.
2. Write a custom inelastic rate code with variable v_esc and an LMC high-velocity tail to compute LZ event counts across delta_m = 400–600 keV and test the 481–519 keV region directly.
3. Replace the tool's p-wave indirect pass with an s-wave WW/ZZ (about 1e-26 cm^3/s) comparison against Fermi dwarf and H.E.S.S. continuum and line limits at 1.1 TeV.
4. Obtain an independent Sommerfeld-plus-coannihilation thermal mass (e.g. via the arXiv:1008.2905 method) and reconcile it with 1.08 ± 0.02 TeV and the inferred 1172 GeV.
5. Re-read narrow windows of arXiv:2609.01590 and arXiv:2609.07807 to validate the contested numbers, including whether the CTAO Omega 0.85 is the paper's own value.
6. Extract the per-treatment IceCube delta limits at 1.1 TeV from arXiv:2610.03692 Fig. 2.
7. Add isotope averaging over natural xenon and the ±23 keV energy uncert